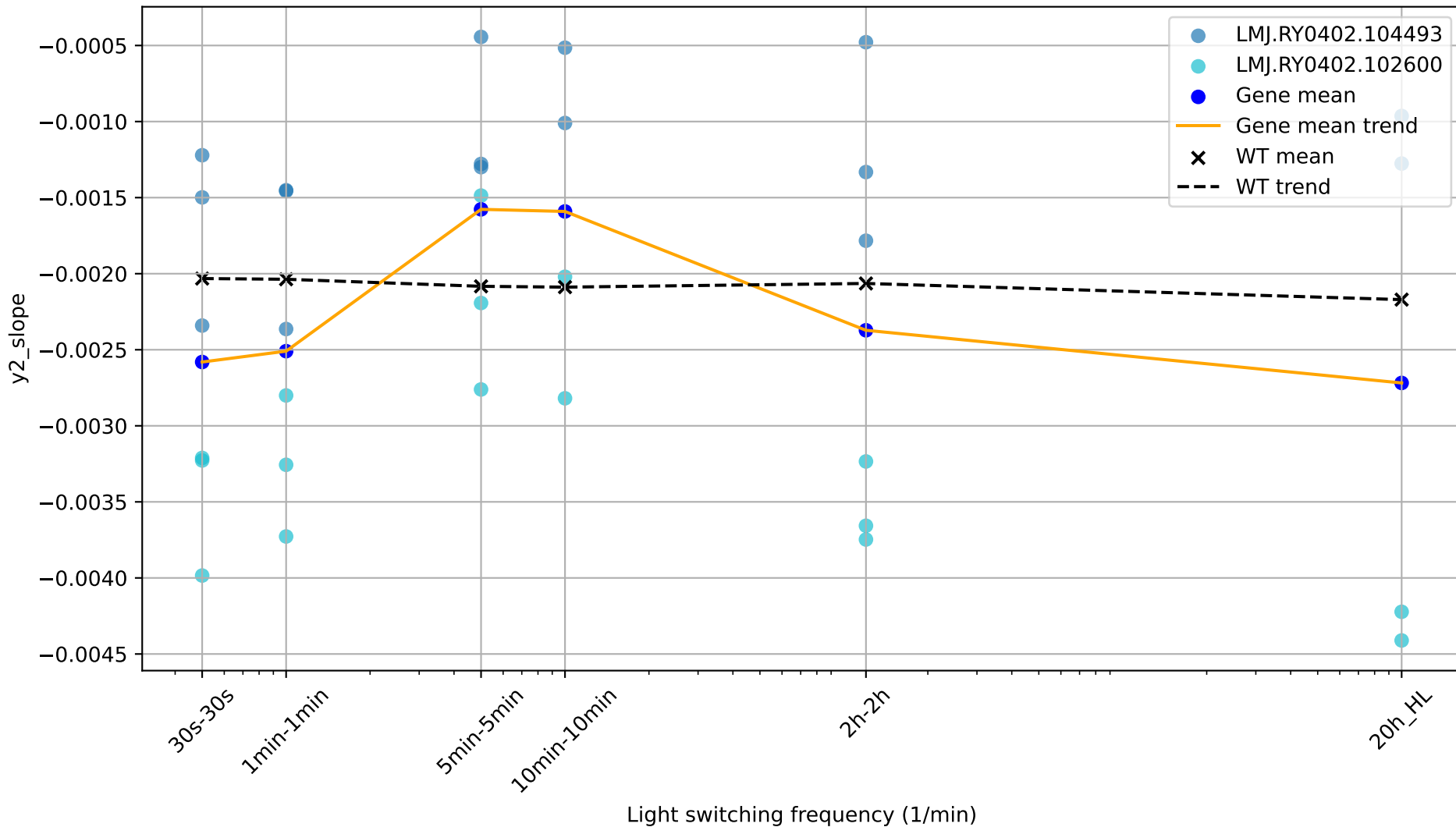
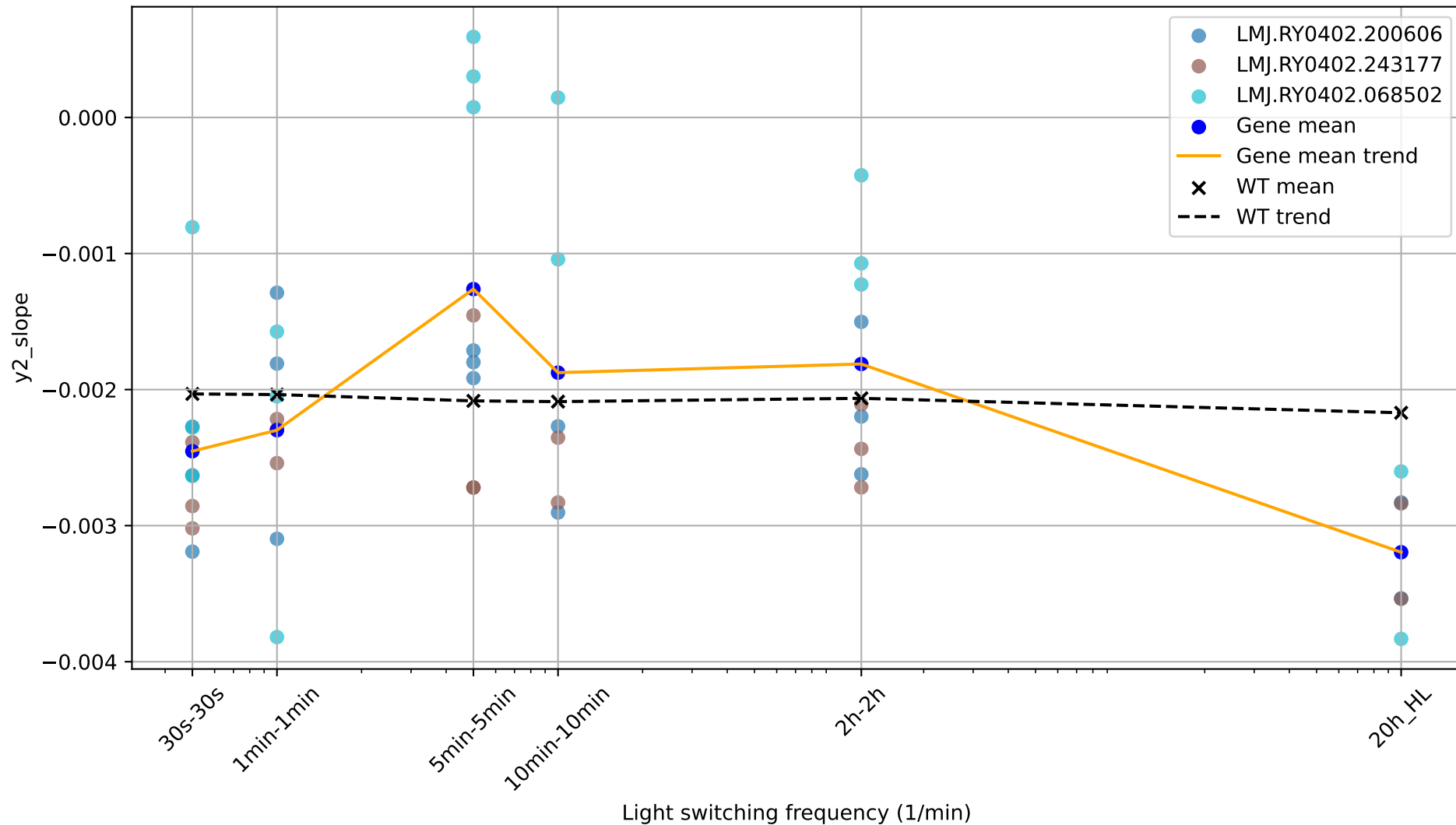


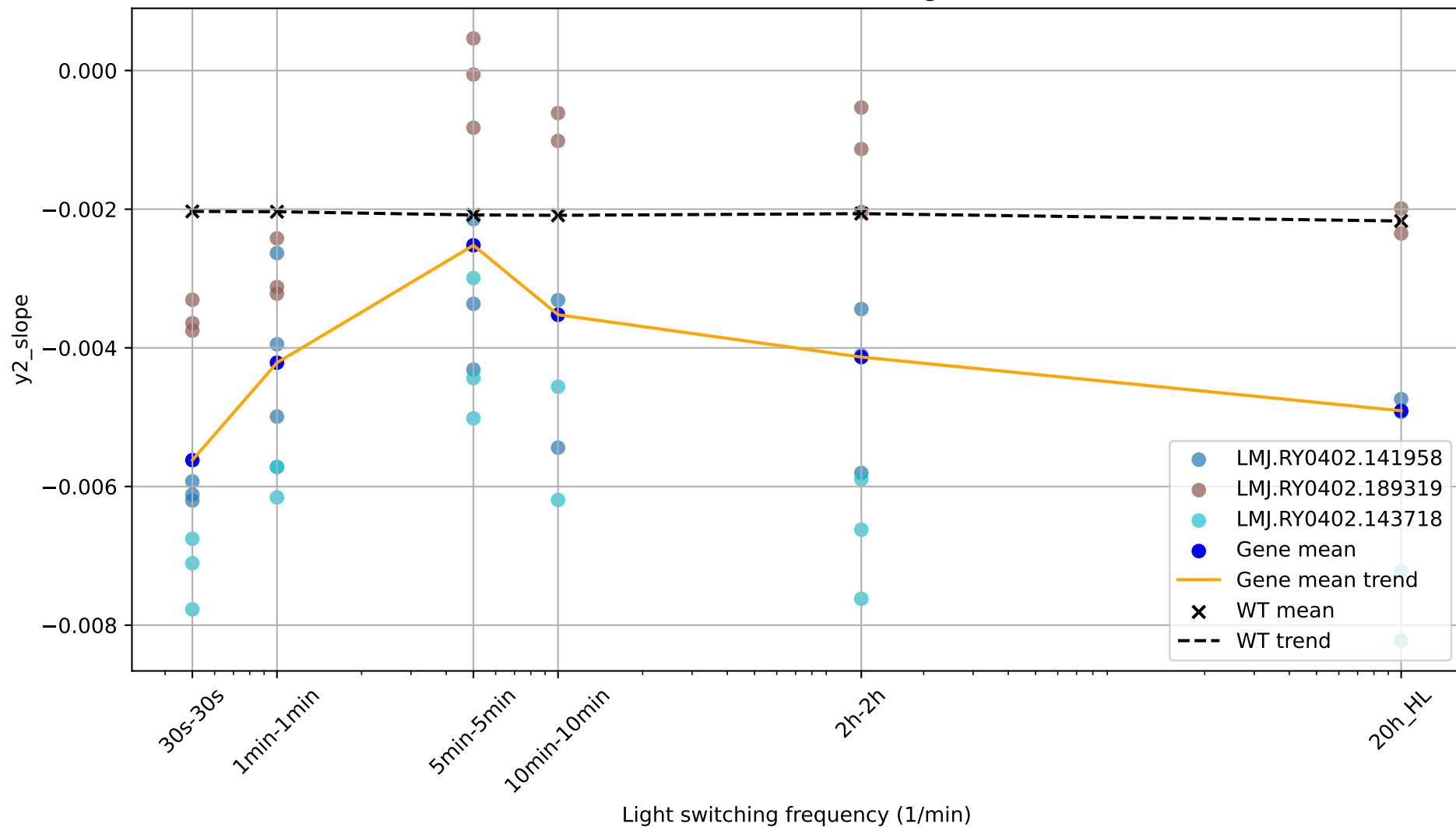
Mutant vs WT - Gene: Cre17.g716400



Mutant vs WT - Gene: Cre17.g709900



Mutant vs WT - Gene: Cre16.g648750



Mutant vs WT - Gene: Cre09.g403800,Cre12.g524700 & Cre12.g524750

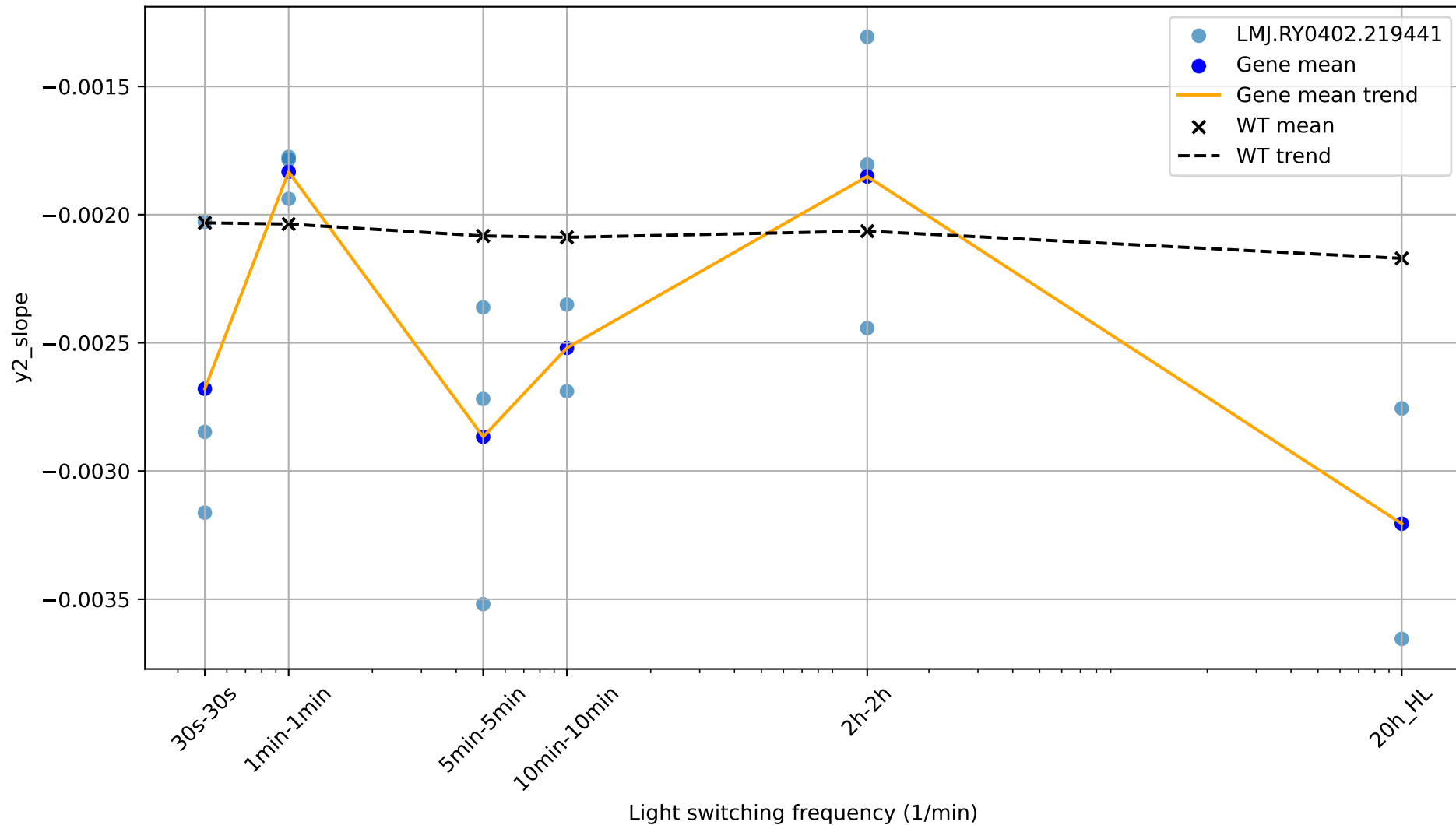


Figure 10: Effect of light switching frequency on gene expression. The graph plots log₂(FPKM) against light switching frequency (1/min). The x-axis categories are 30s-30s, 1min-1min, 5min-5min, 10min-10min, 2h-2h, and 20h_HL. The y-axis ranges from 0 to 12. The legend indicates: LMJ.RY0402.046272 (light blue dots), Gene mean (blue dots), Gene mean trend (orange line), WT mean (black 'x'), and WT trend (dashed black line). The gene mean expression peaks at 5min-5min and then decreases, while the WT mean remains relatively constant.

Light switching frequency (1/min)	LMJ.RY0402.046272 (log ₂ (FPKM))	Gene mean (log ₂ (FPKM))	WT mean (log ₂ (FPKM))
30s-30s	~1.5, ~2.5, ~3.5, ~4.5	~3.5	~11.5
1min-1min	~3.5, ~4.5, ~5.5	~5.5	~11.5
5min-5min	~5.5, ~6.5, ~7.5, ~8.5	~8.5	~11.5
10min-10min	~4.5, ~5.5, ~6.5	~5.5	~11.5
2h-2h	~1.5, ~2.5, ~3.5	~2.5	~11.5
20h_HL	~1.5, ~2.5, ~3.5	~2.5	~11.5

1min-1min

5min-5min

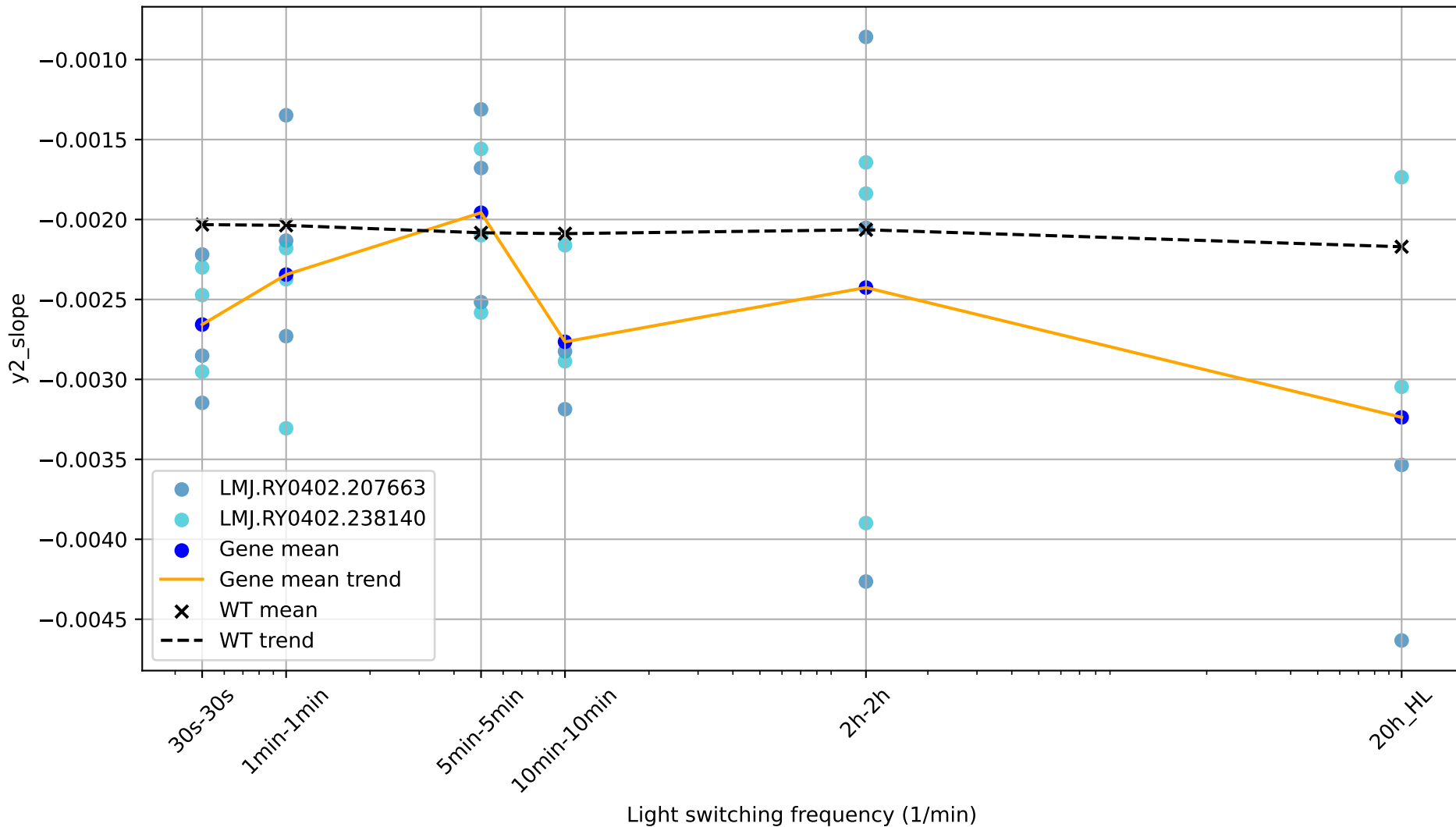
10min-10min

2h-2h

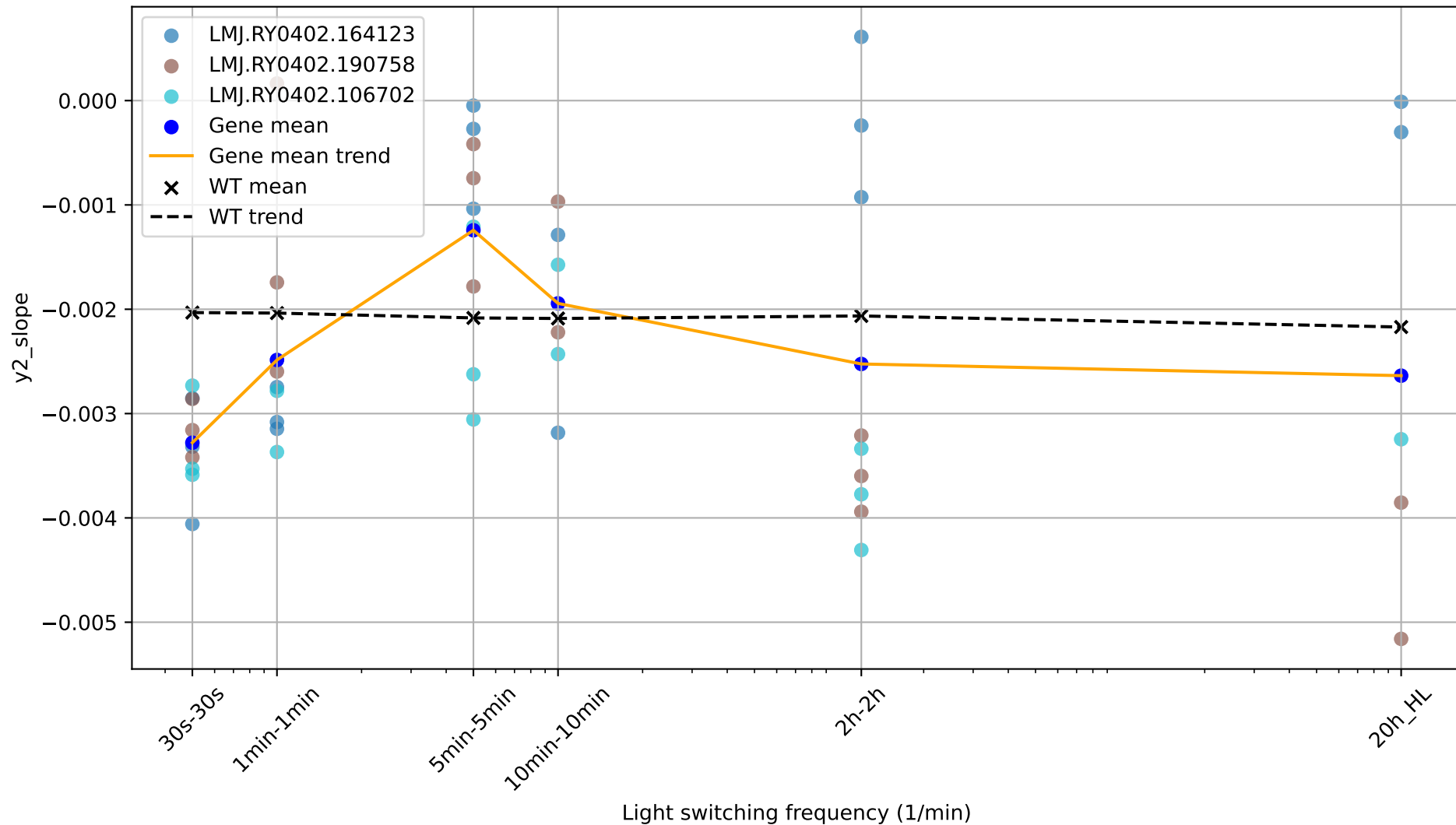
20h_HL

Light switching frequency (1/min)

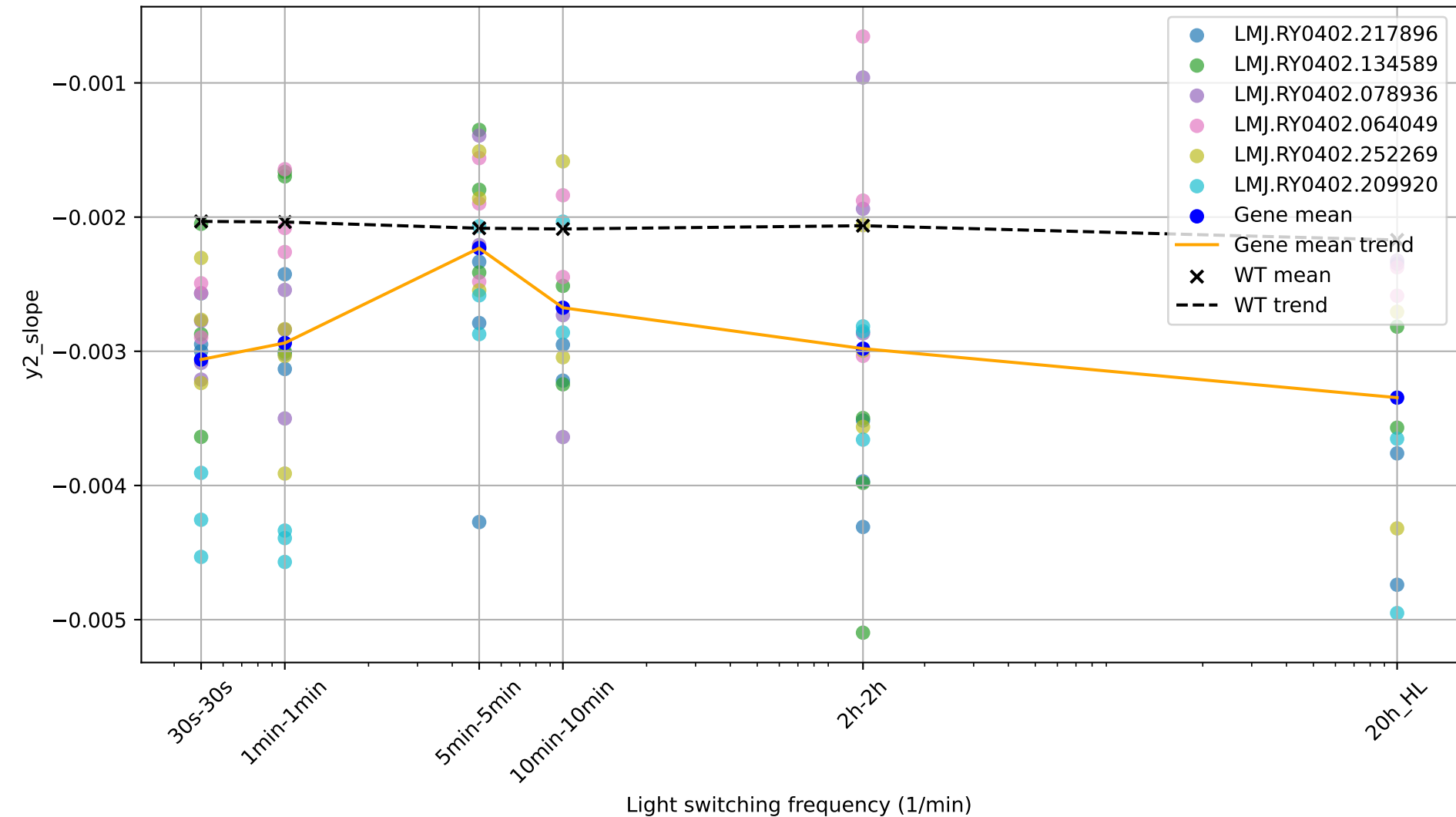
Mutant vs WT - Gene: Cre10.g465450



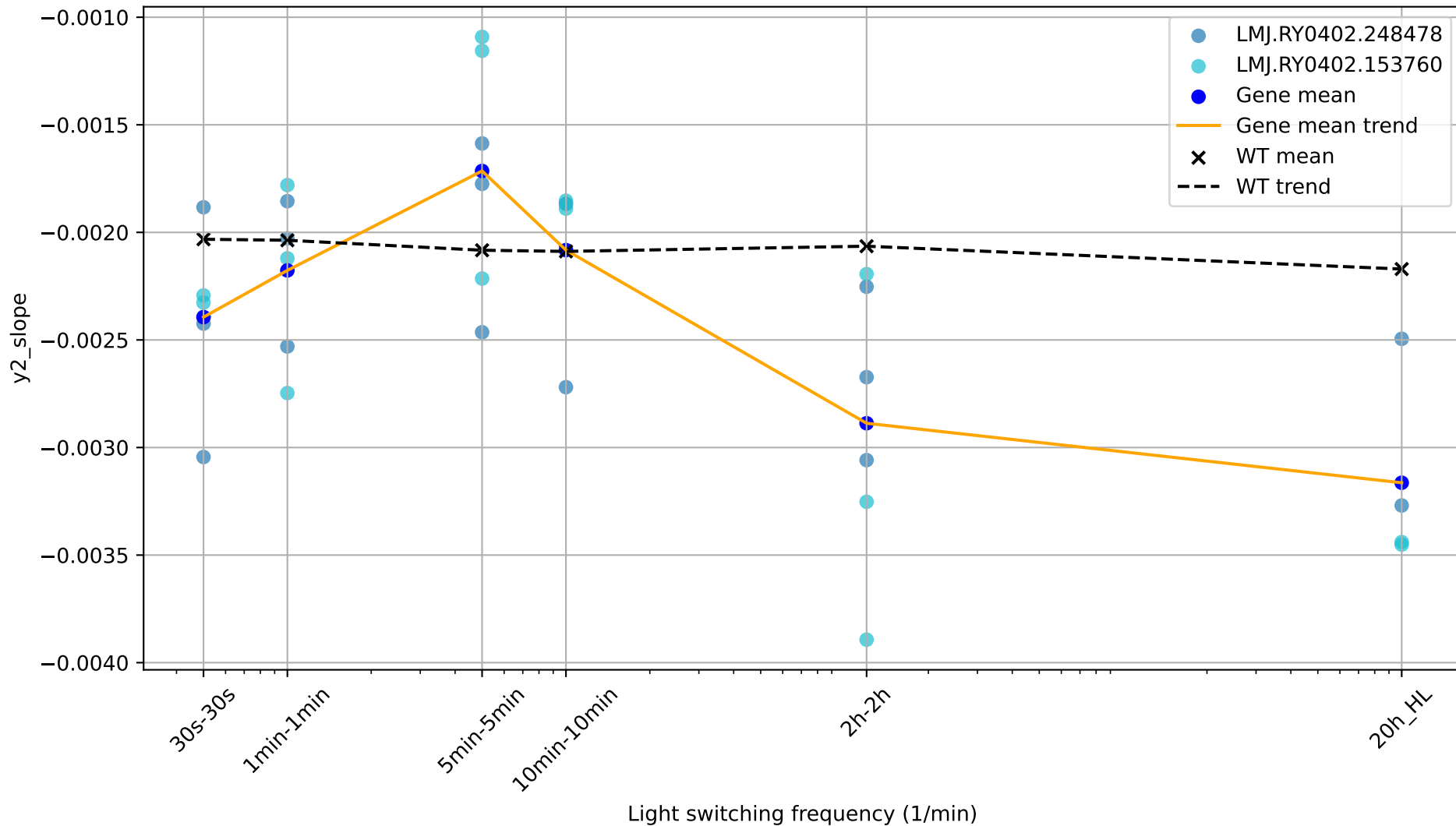
Mutant vs WT - Gene: Cre01.g040950



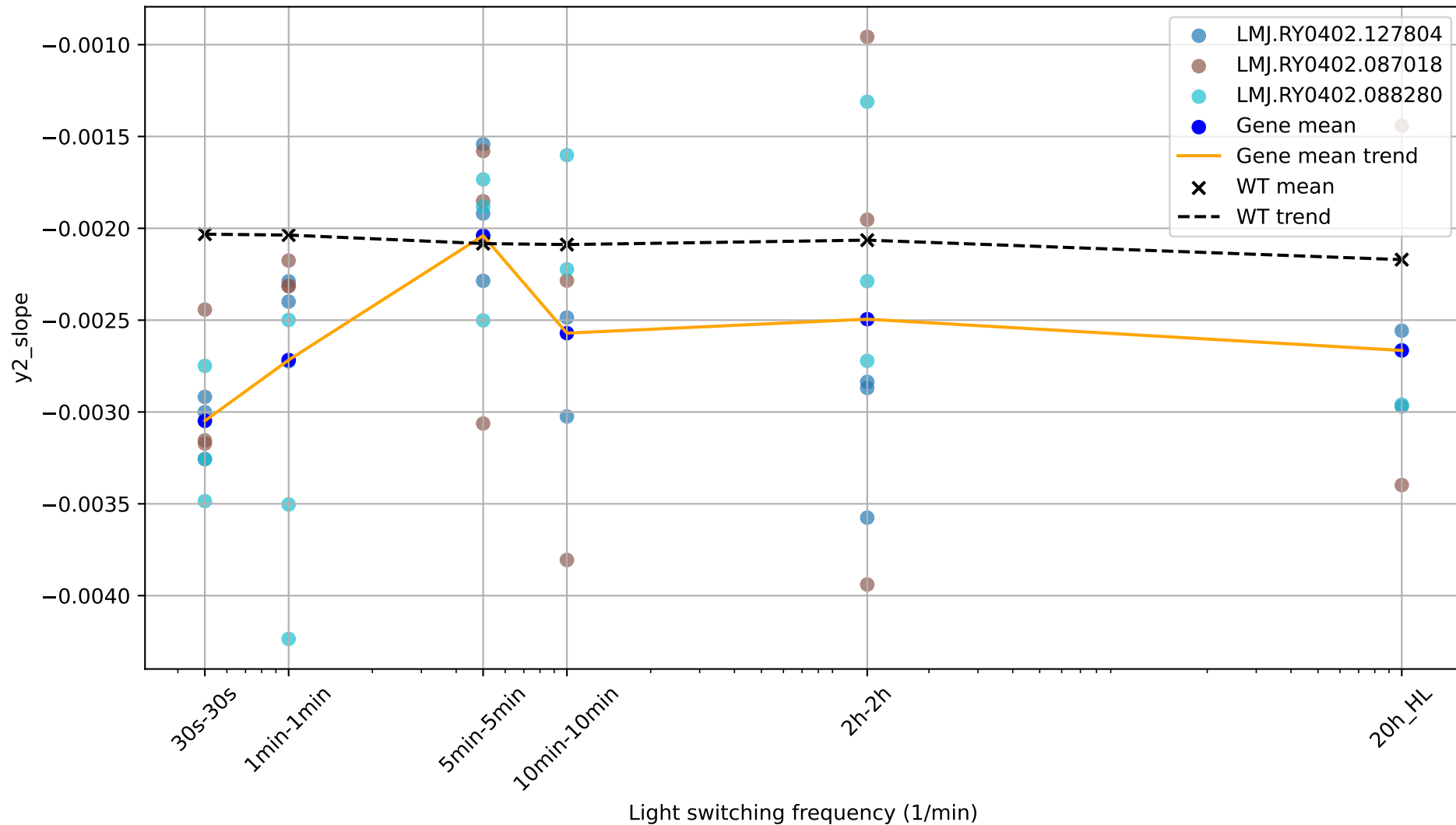
Mutant vs WT - Gene: Cre10.g444000



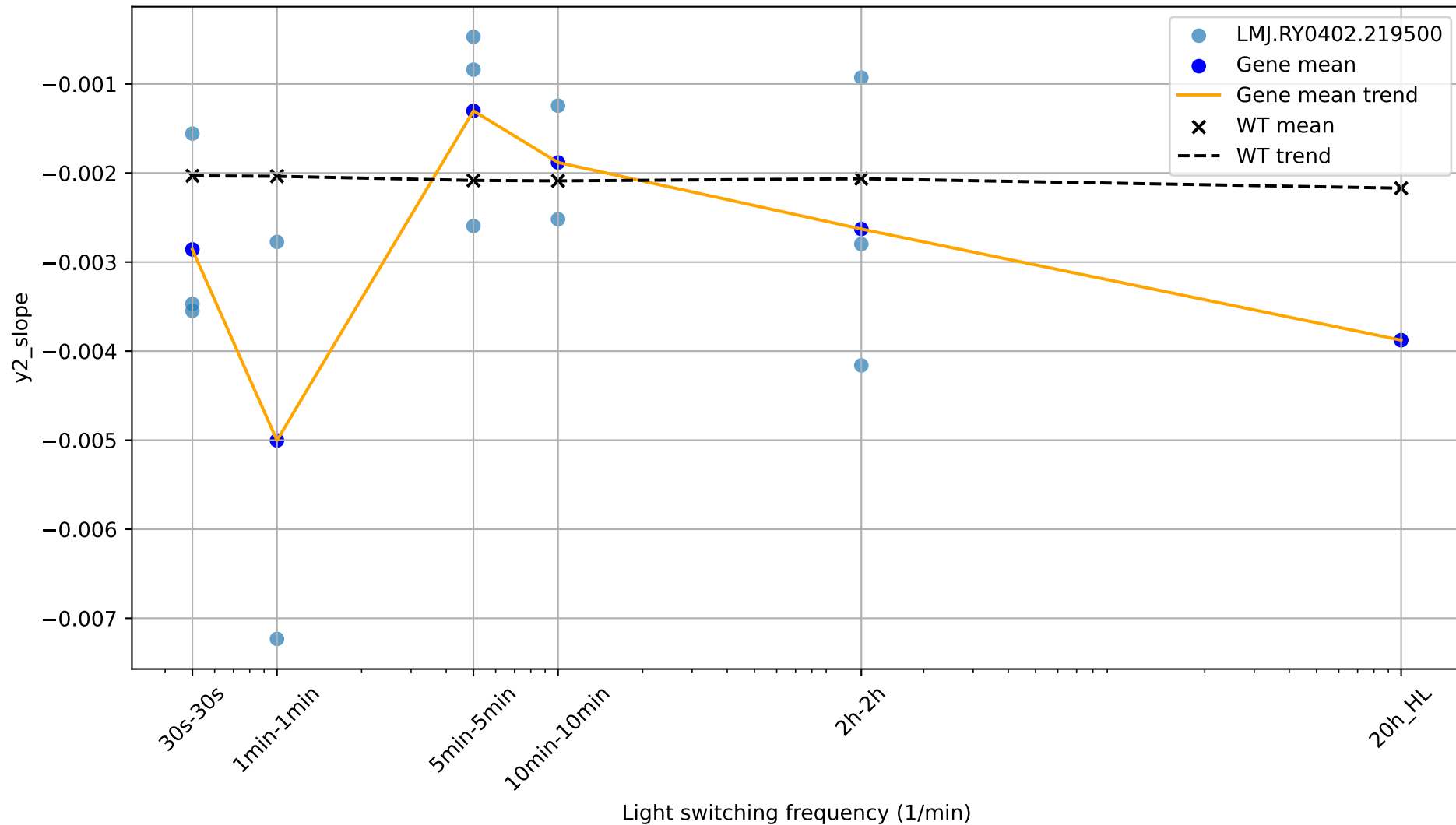
Mutant vs WT - Gene: Cre12.g552600



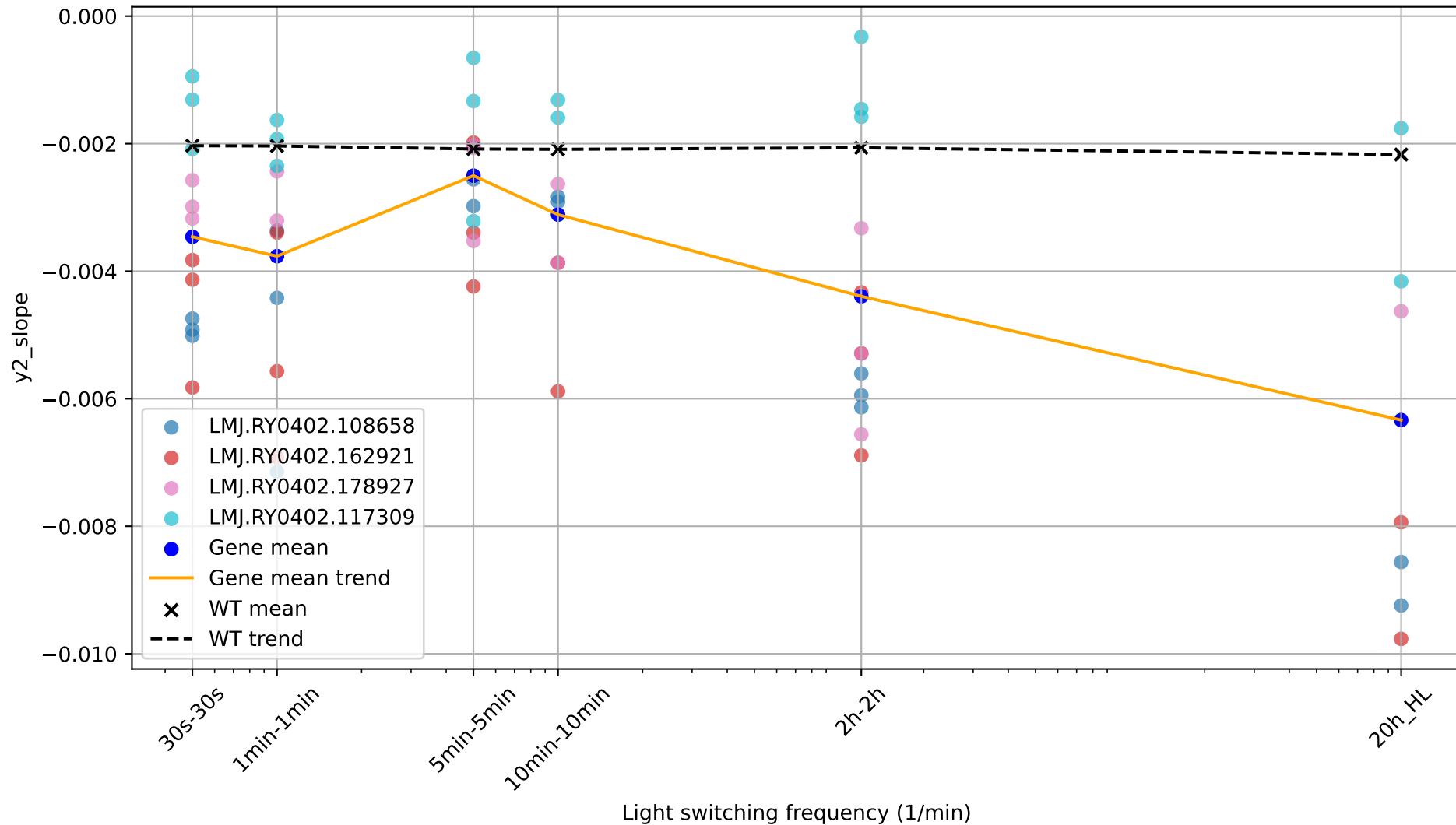
Mutant vs WT - Gene: Cre13.g603700



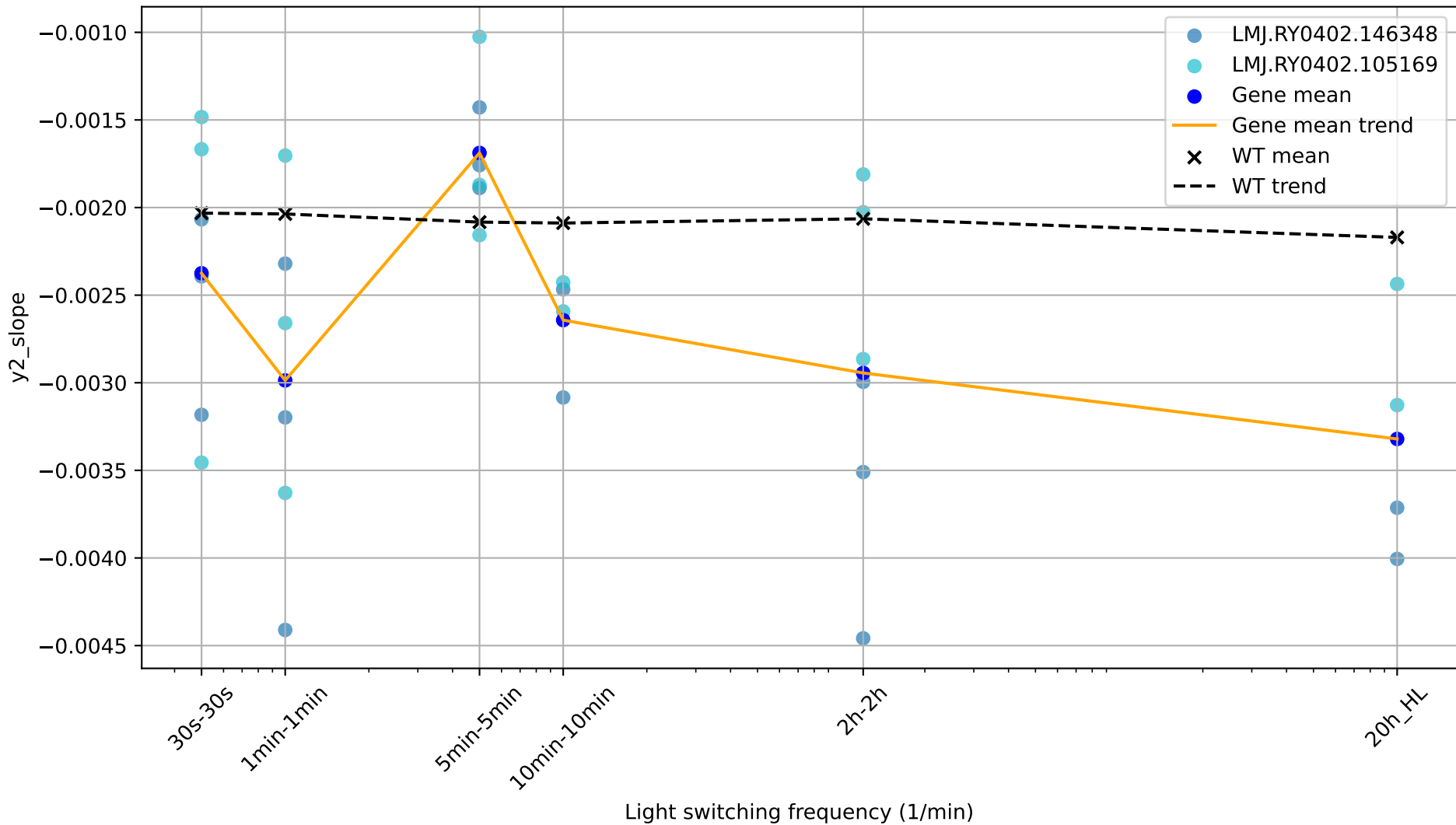
Mutant vs WT - Gene: Cre13.g587850,Cre04.g216850



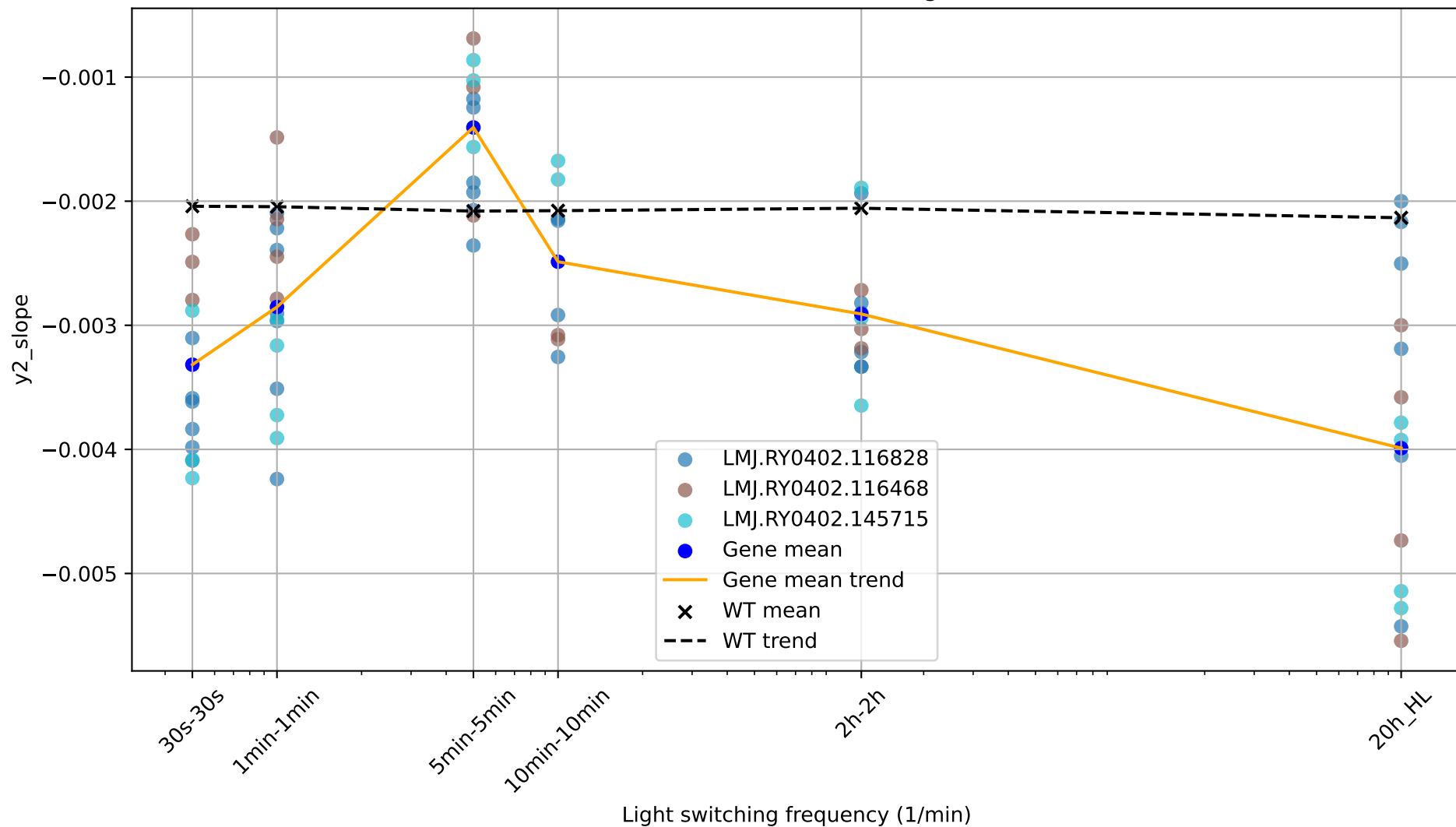
Mutant vs WT - Gene: Cre09.g387949



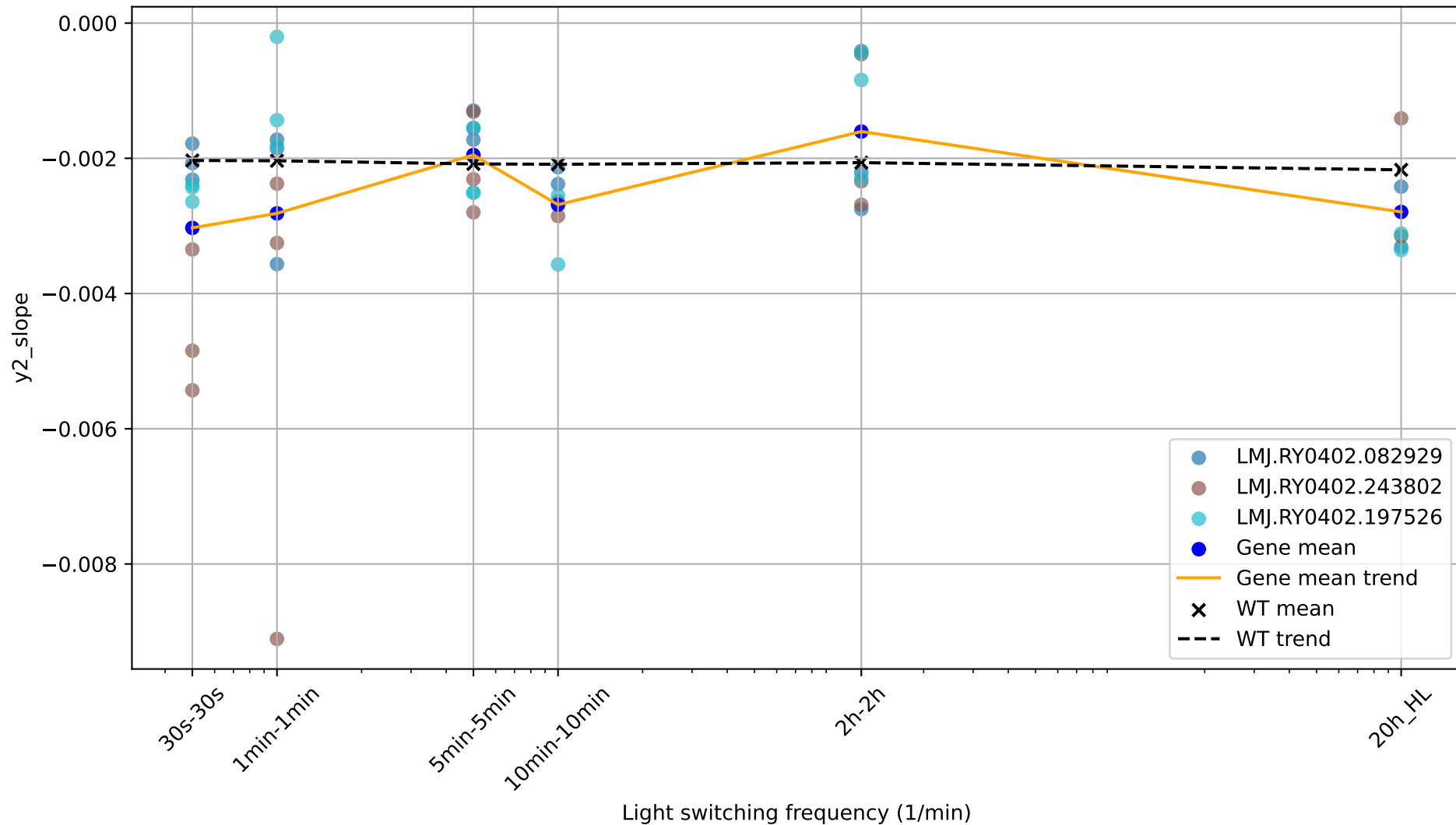
Mutant vs WT - Gene: Cre16.g668250



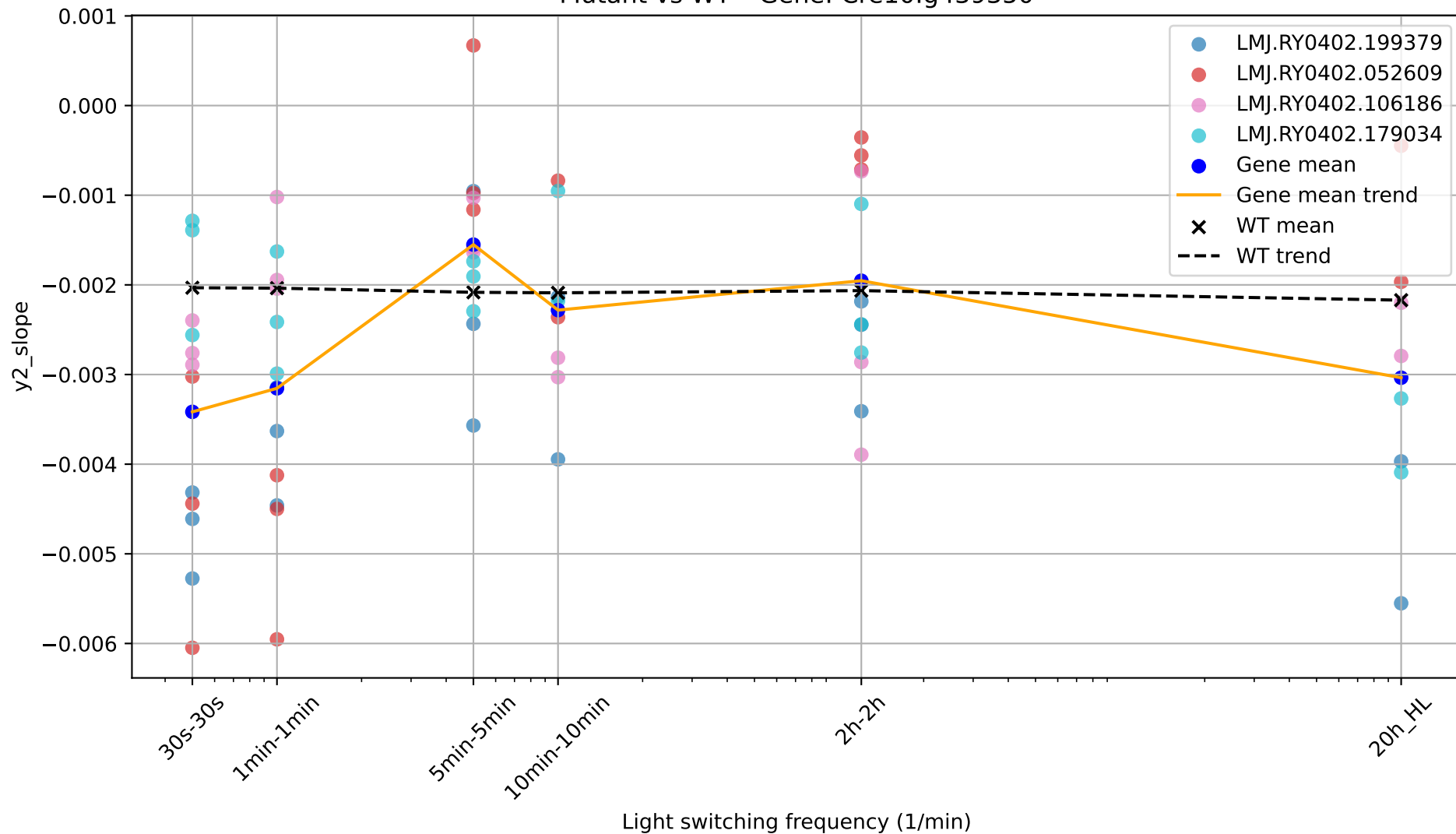
Mutant vs WT - Gene: Cre06.g295600



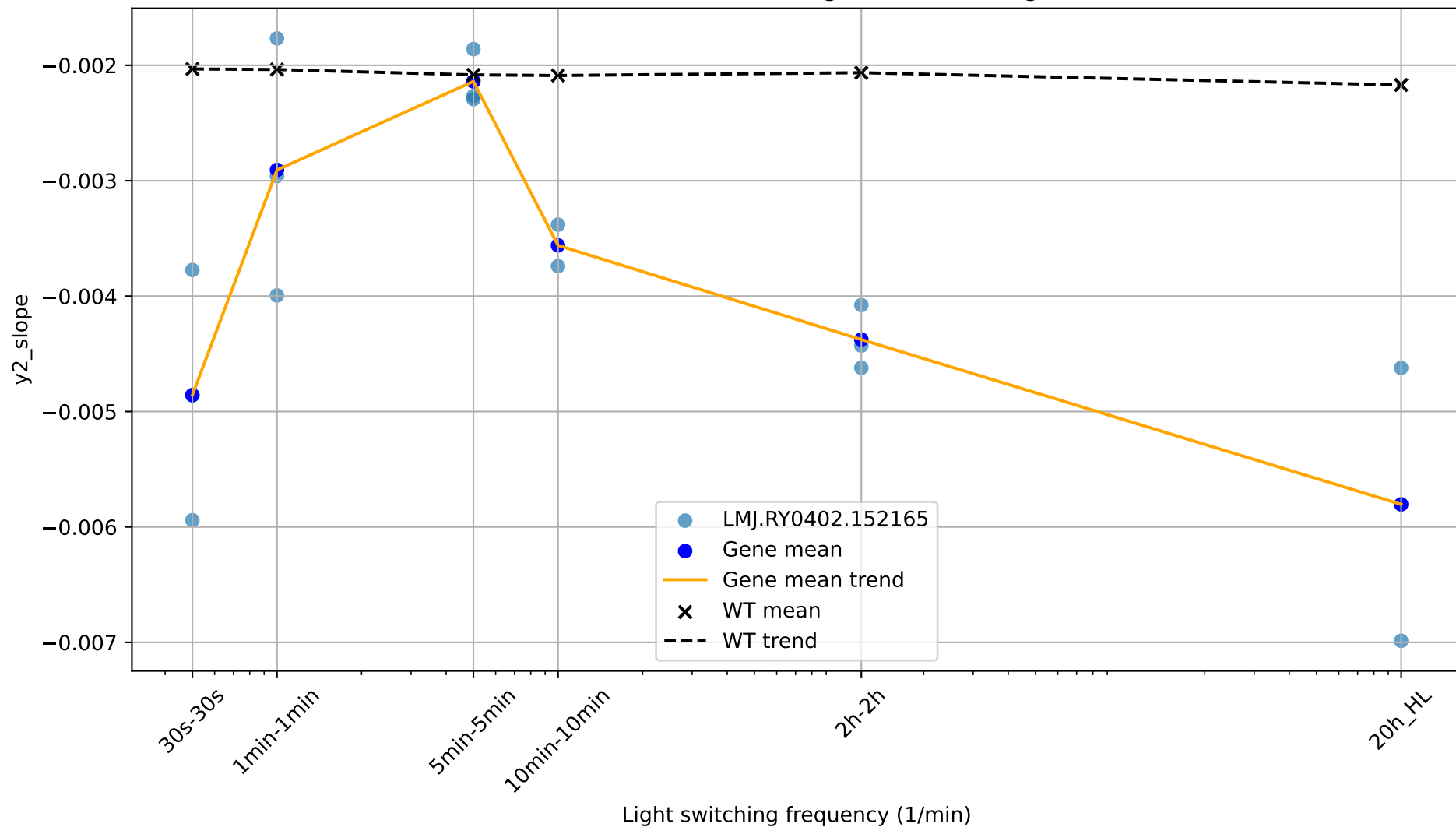
Mutant vs WT - Gene: Cre09.g394695



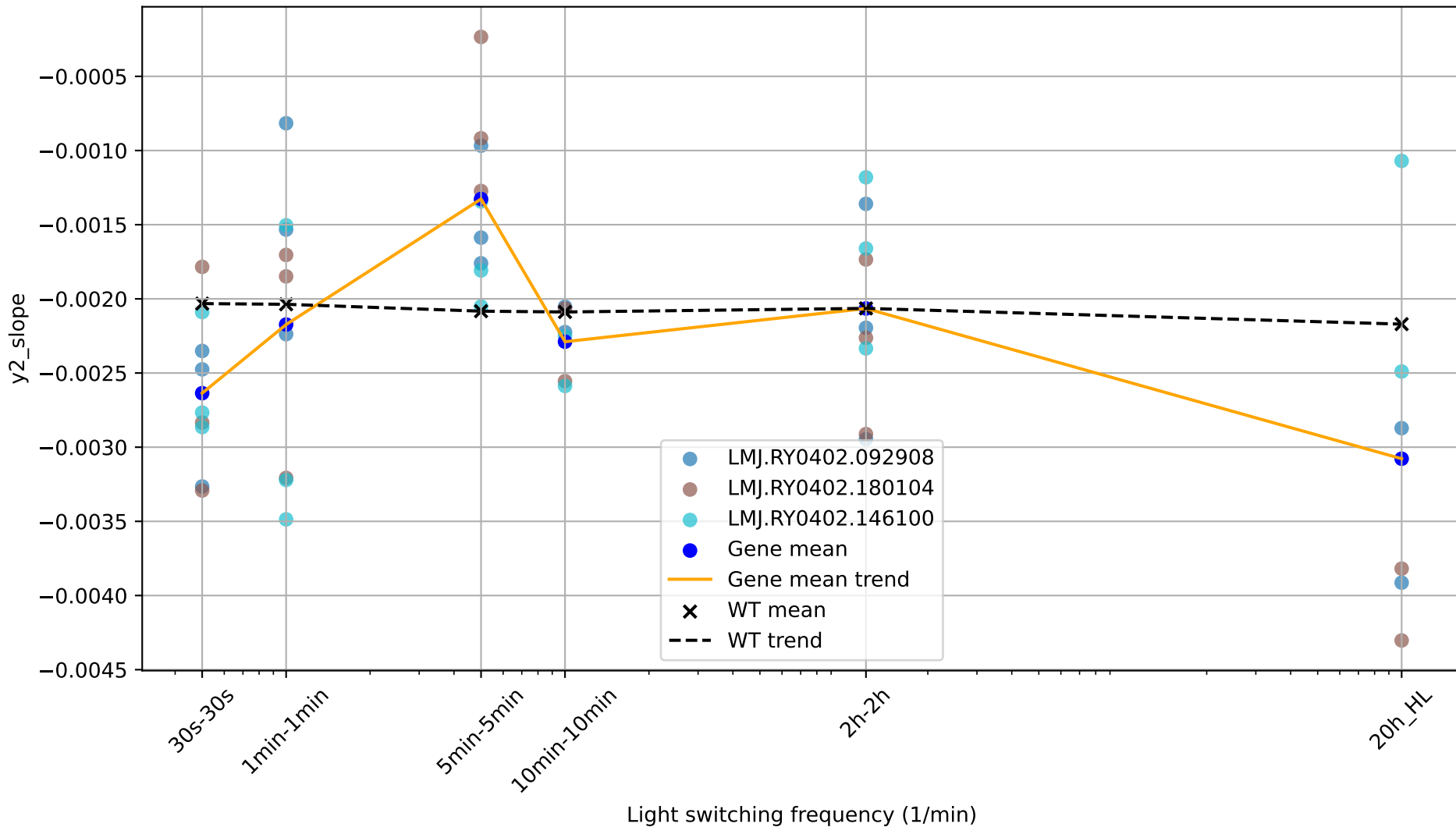
Mutant vs WT - Gene: Cre10.g439350



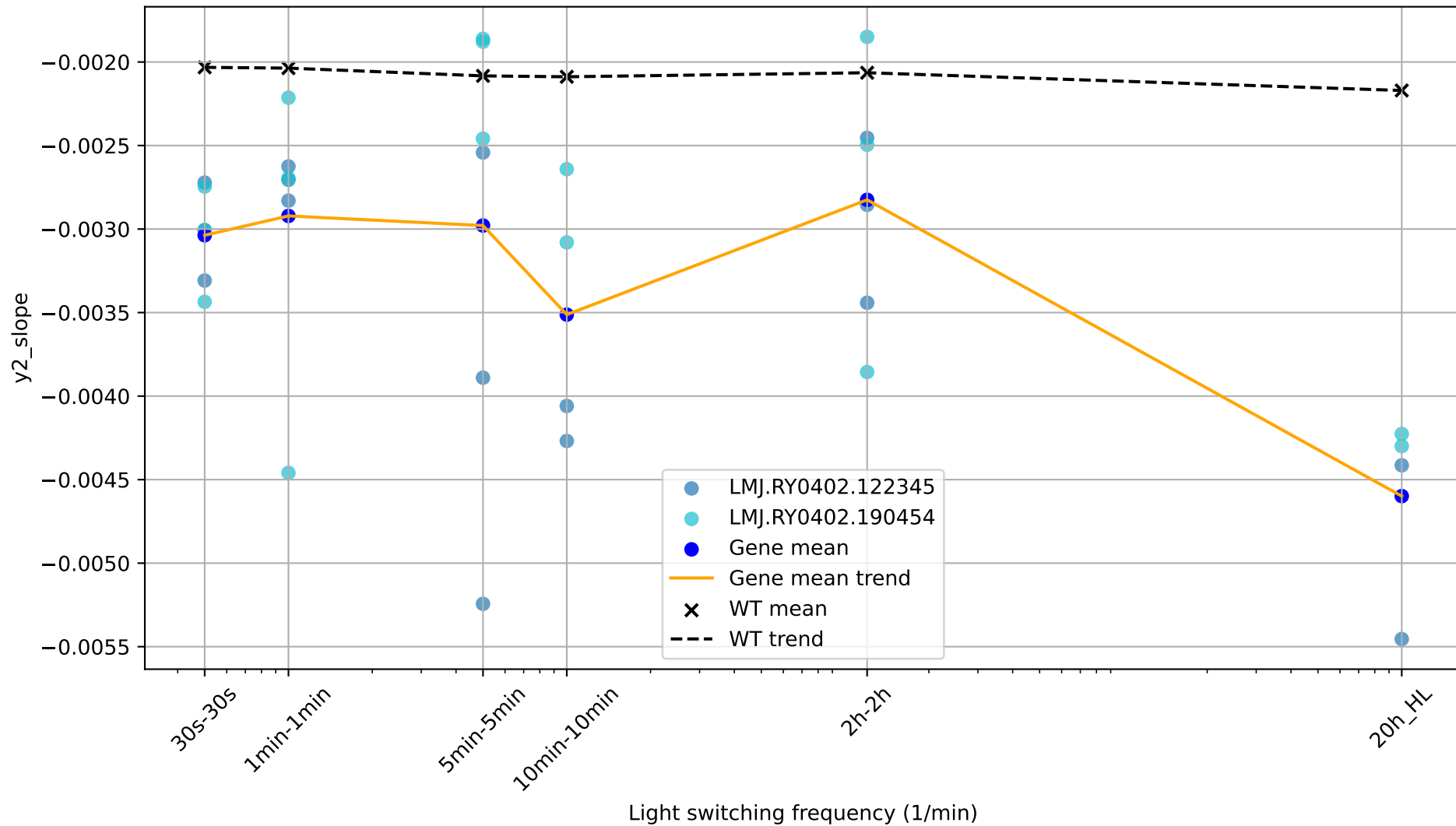
Mutant vs WT - Gene: Cre07.g352250,Cre05.g243802



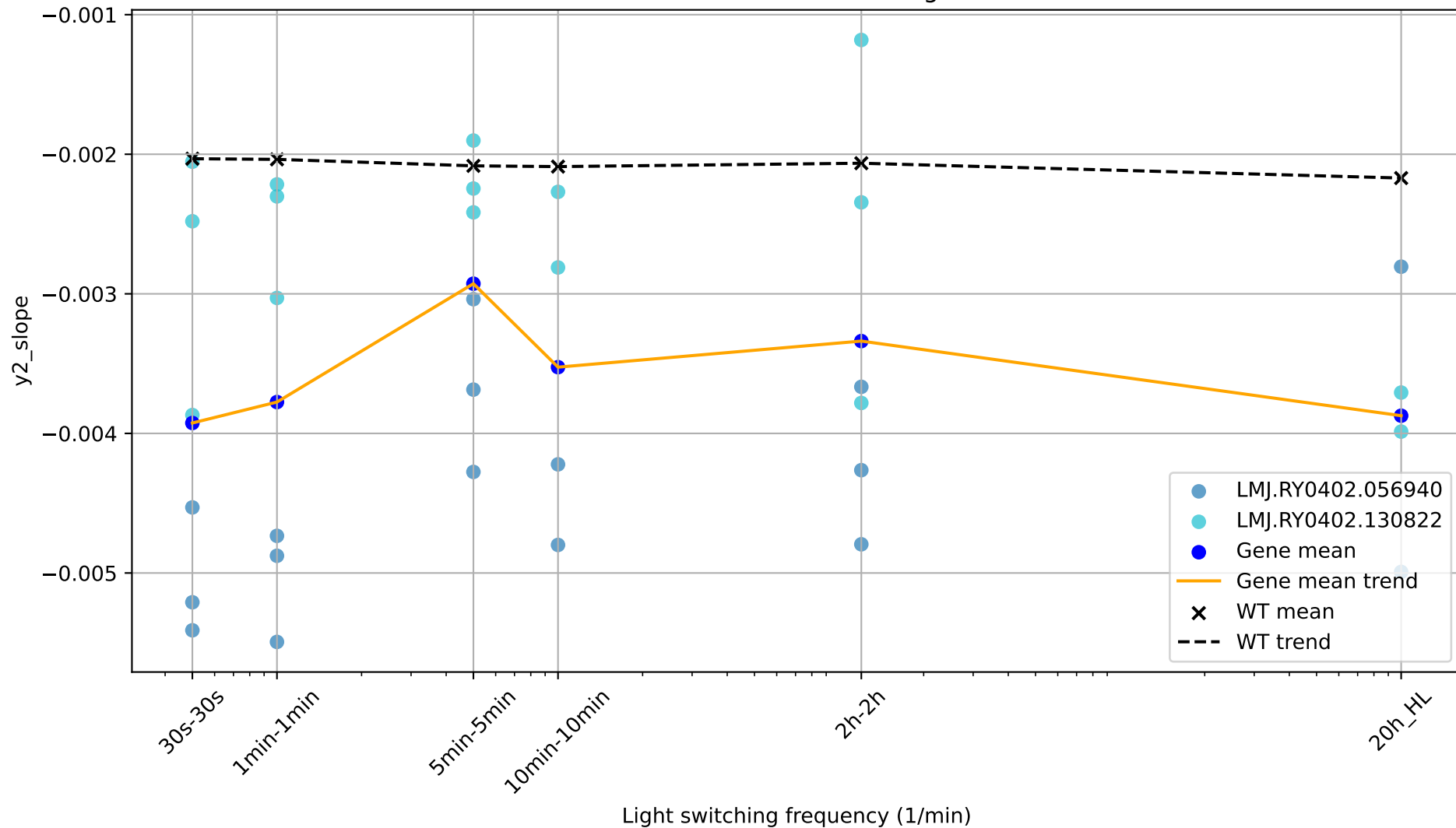
Mutant vs WT - Gene: Cre13.g584901



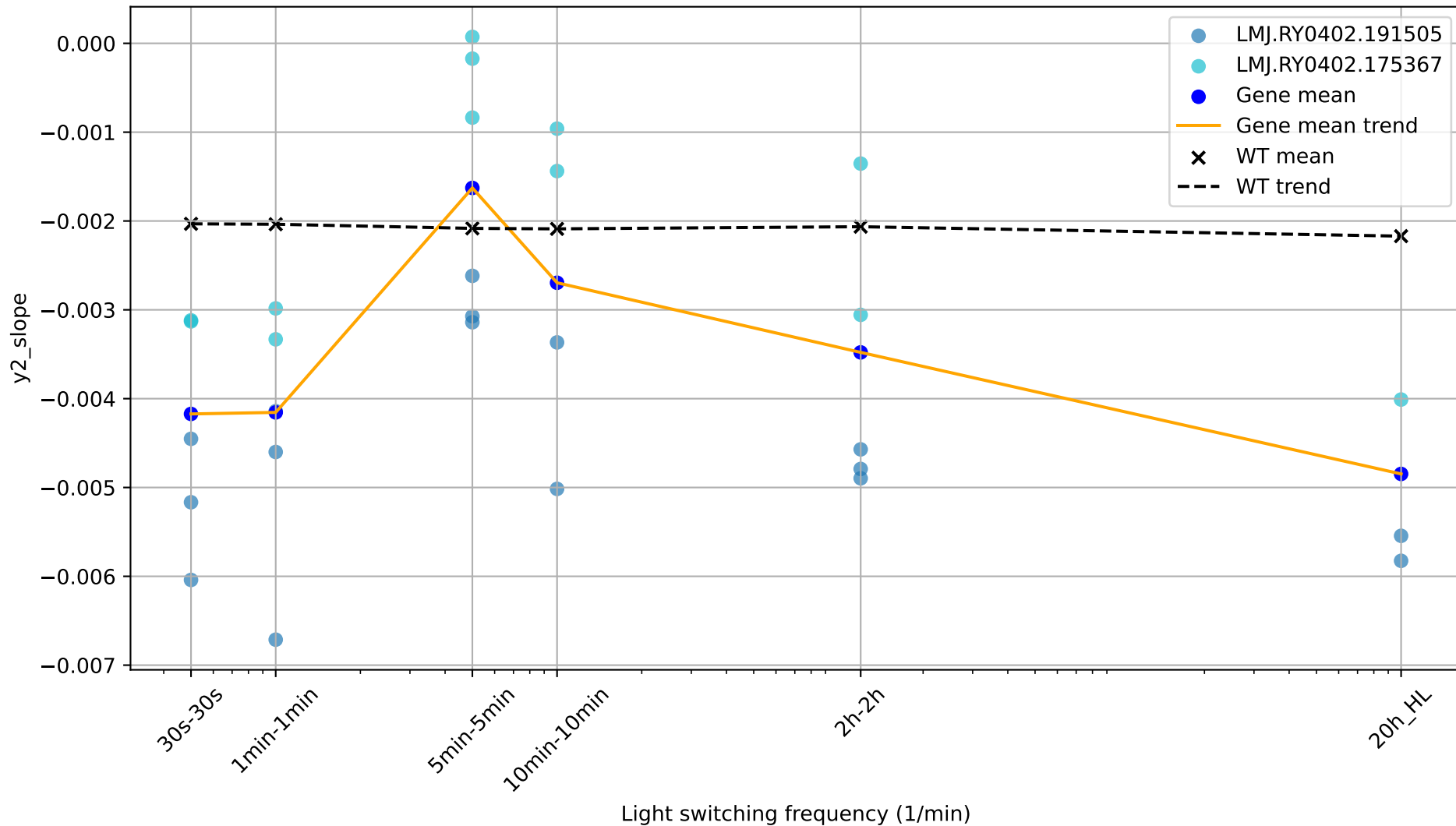
Mutant vs WT - Gene: Cre17.g738050



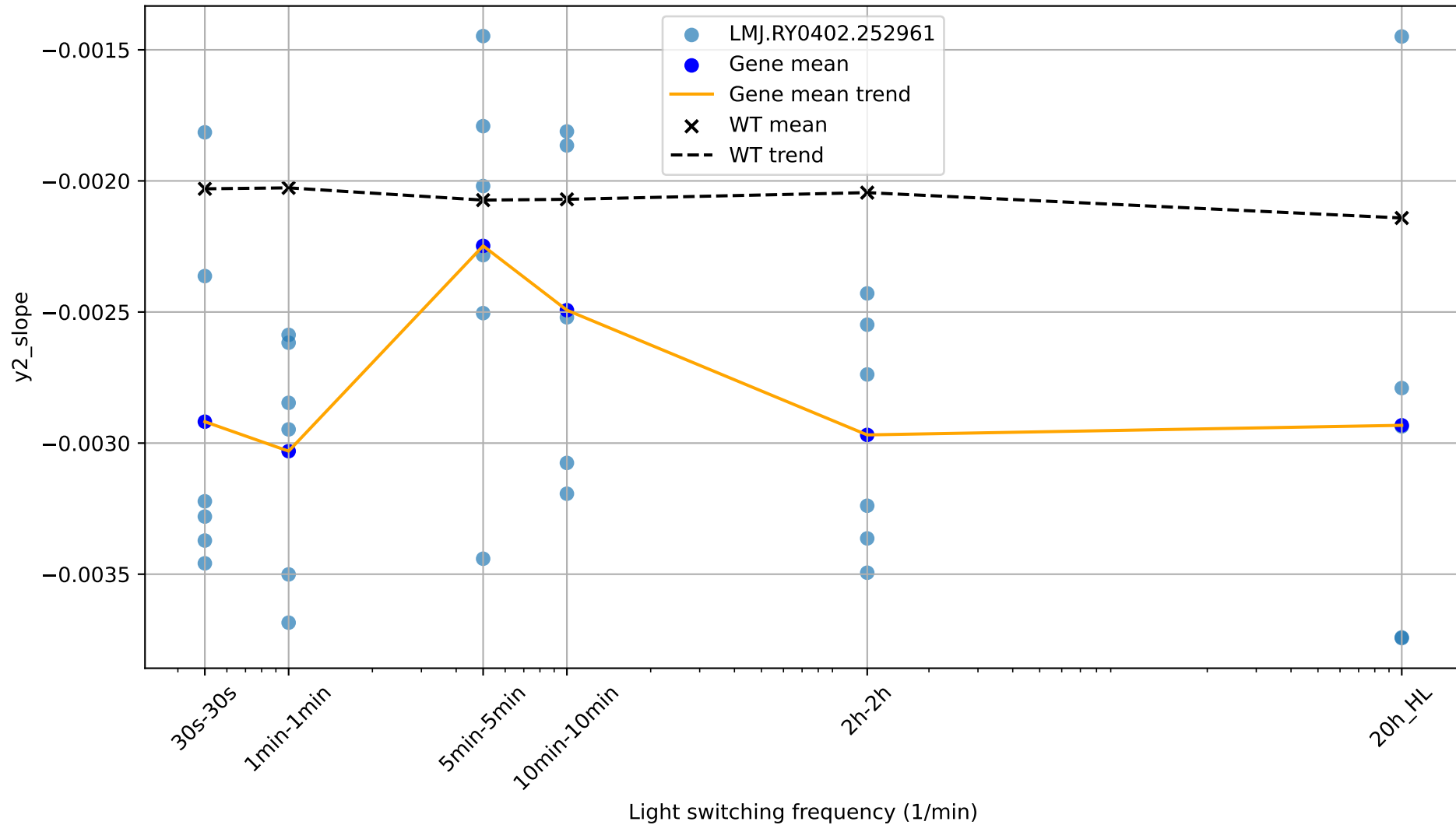
Mutant vs WT - Gene: Cre17.g723950



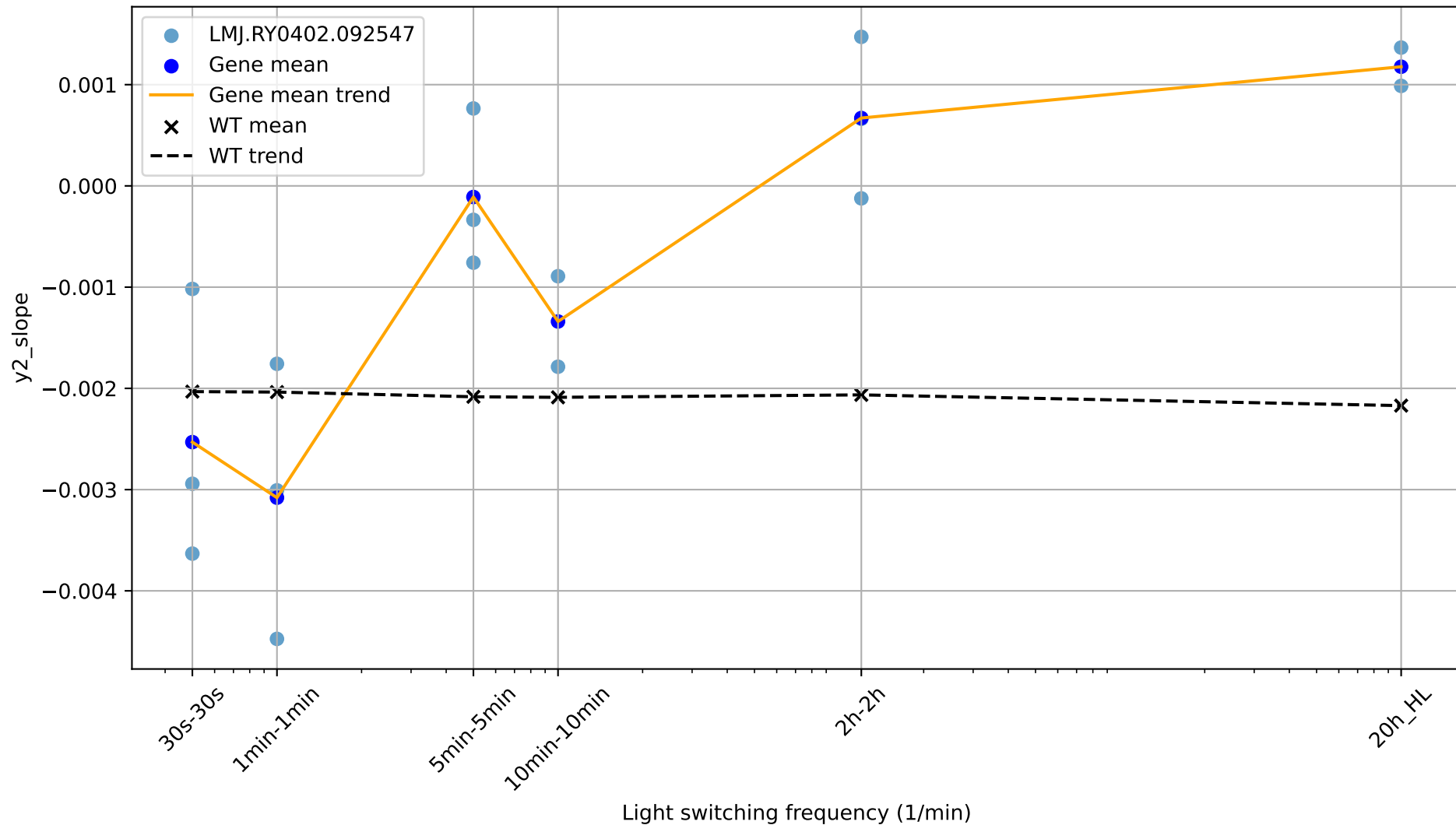
Mutant vs WT - Gene: Cre17.g710300



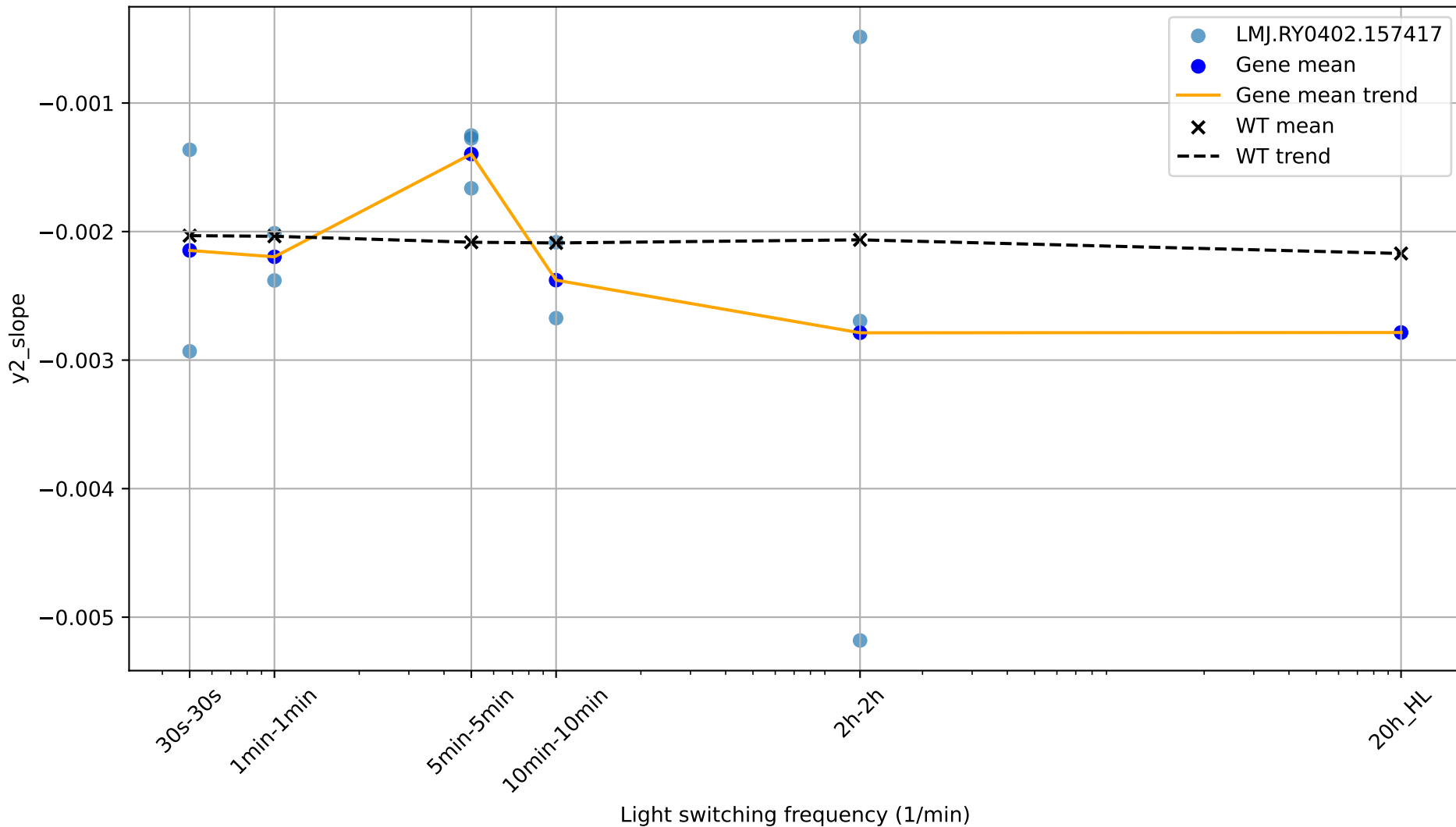
Mutant vs WT - Gene: Cre02.g105150,Cre16.g688638



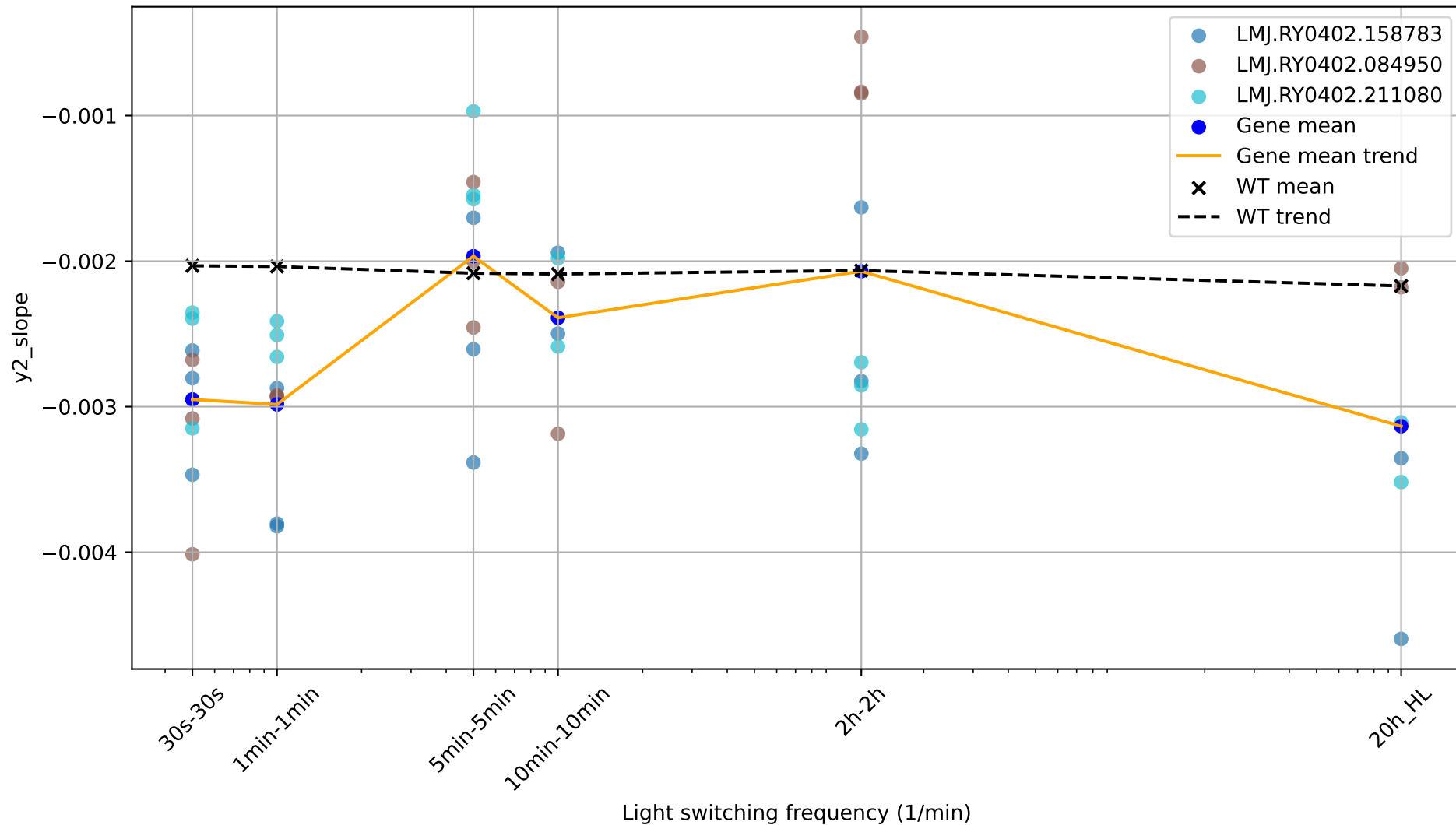
Mutant vs WT - Gene: Cre11.g467741,Cre10.g436150 & Cre10.g436200



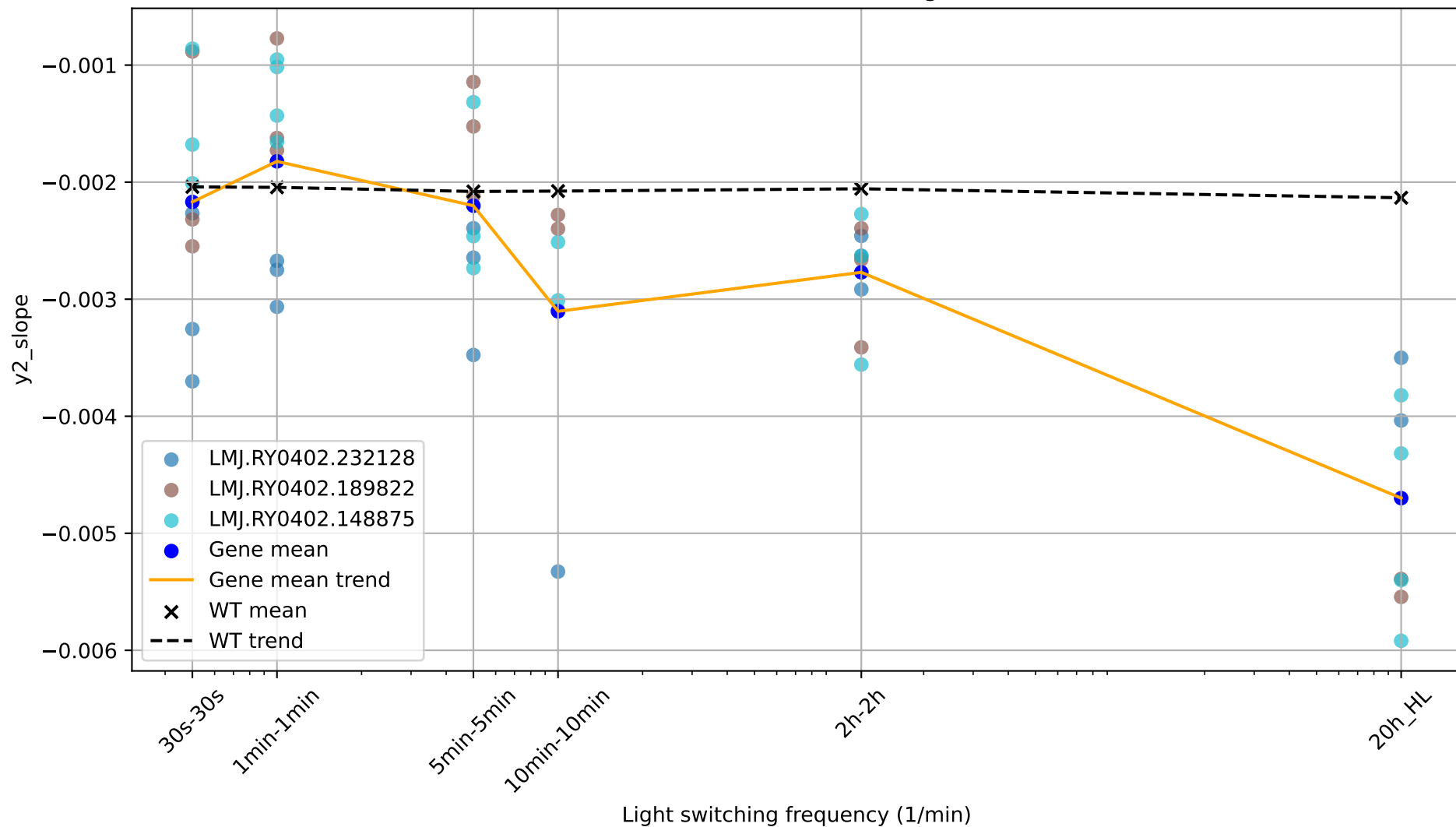
Mutant vs WT - Gene: Cre06.g278119 & Cre06.g278120,Cre05.g236100



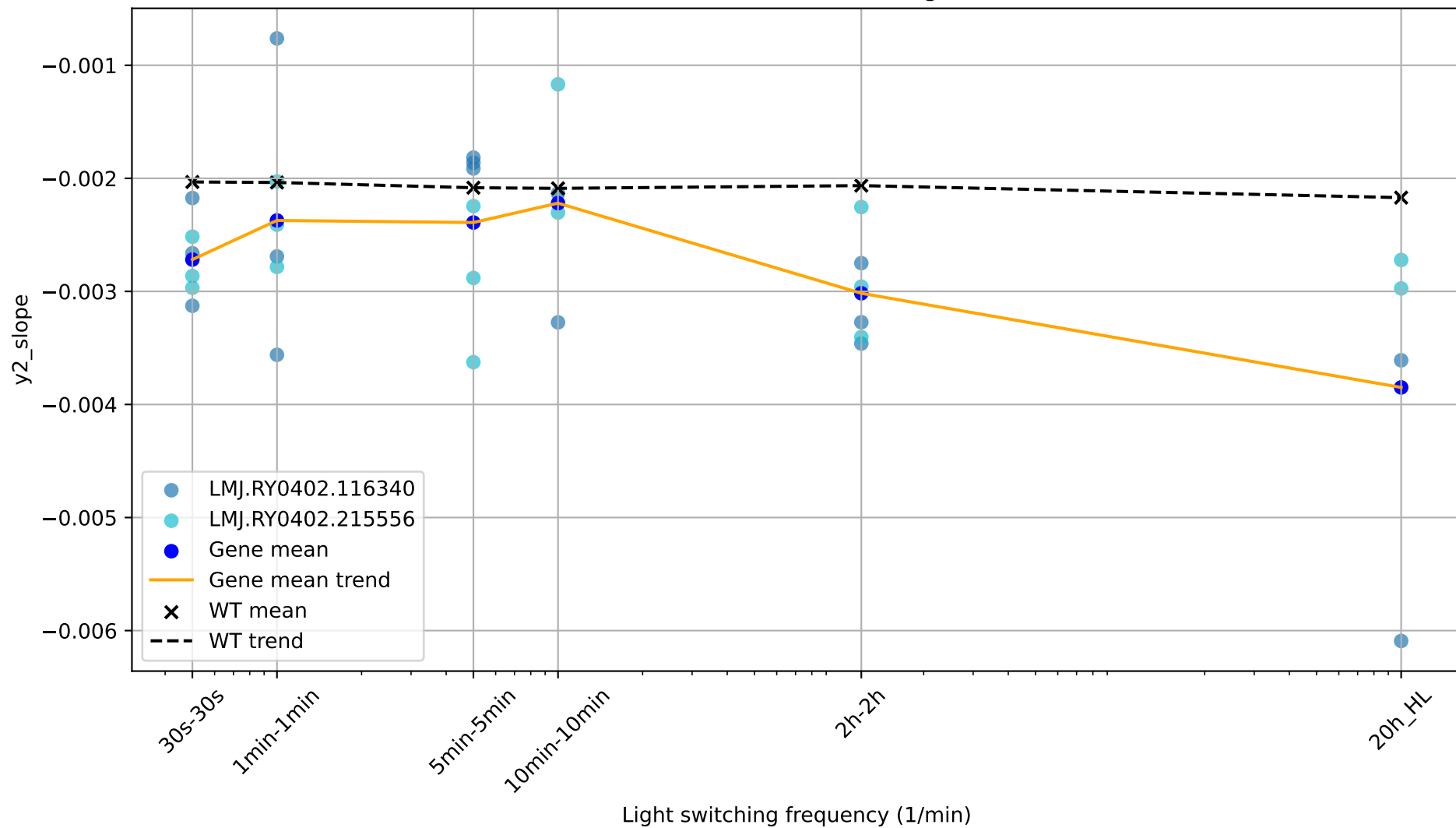
Mutant vs WT - Gene: Cre15.g640750



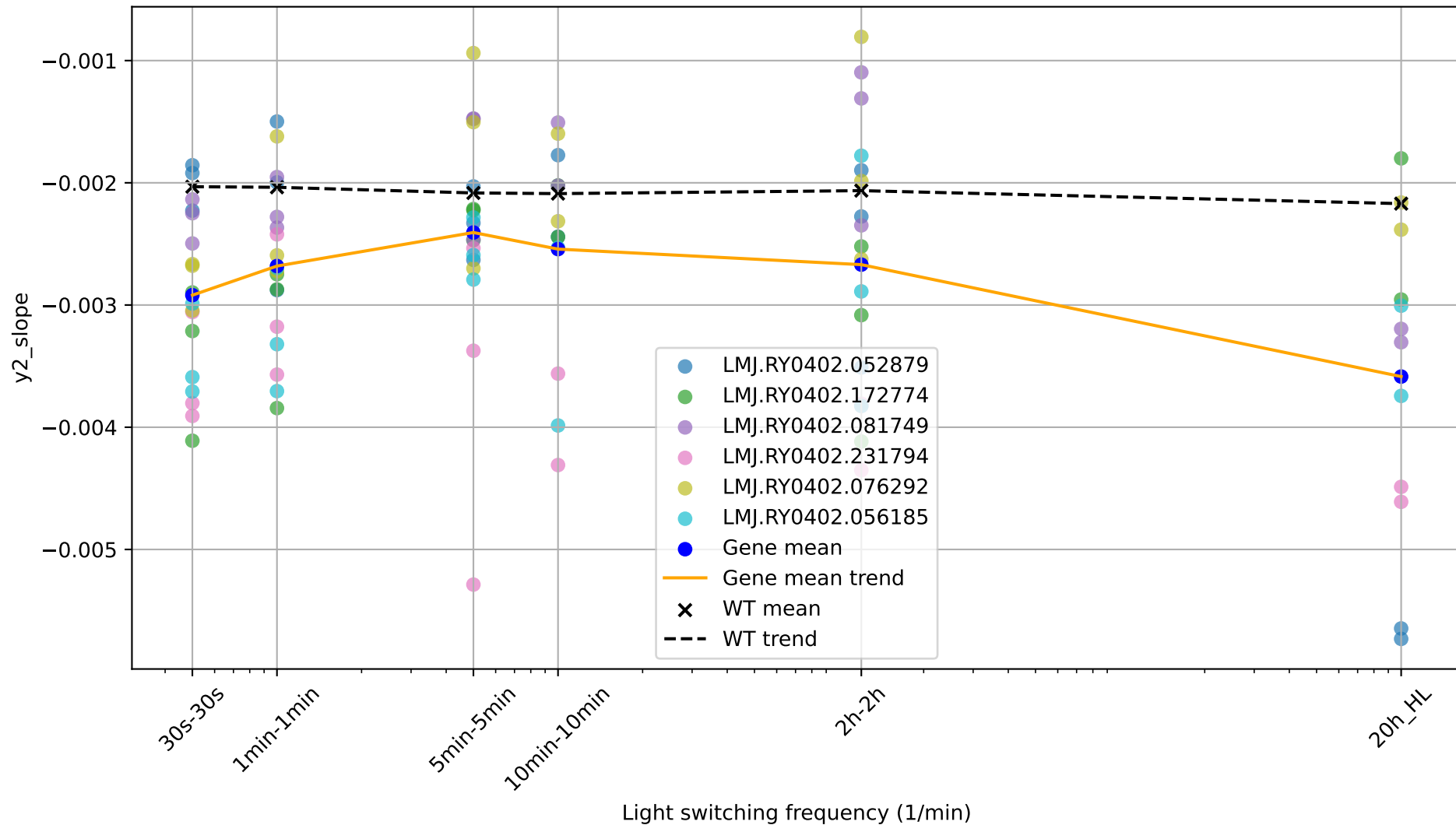
Mutant vs WT - Gene: Cre12.g503500



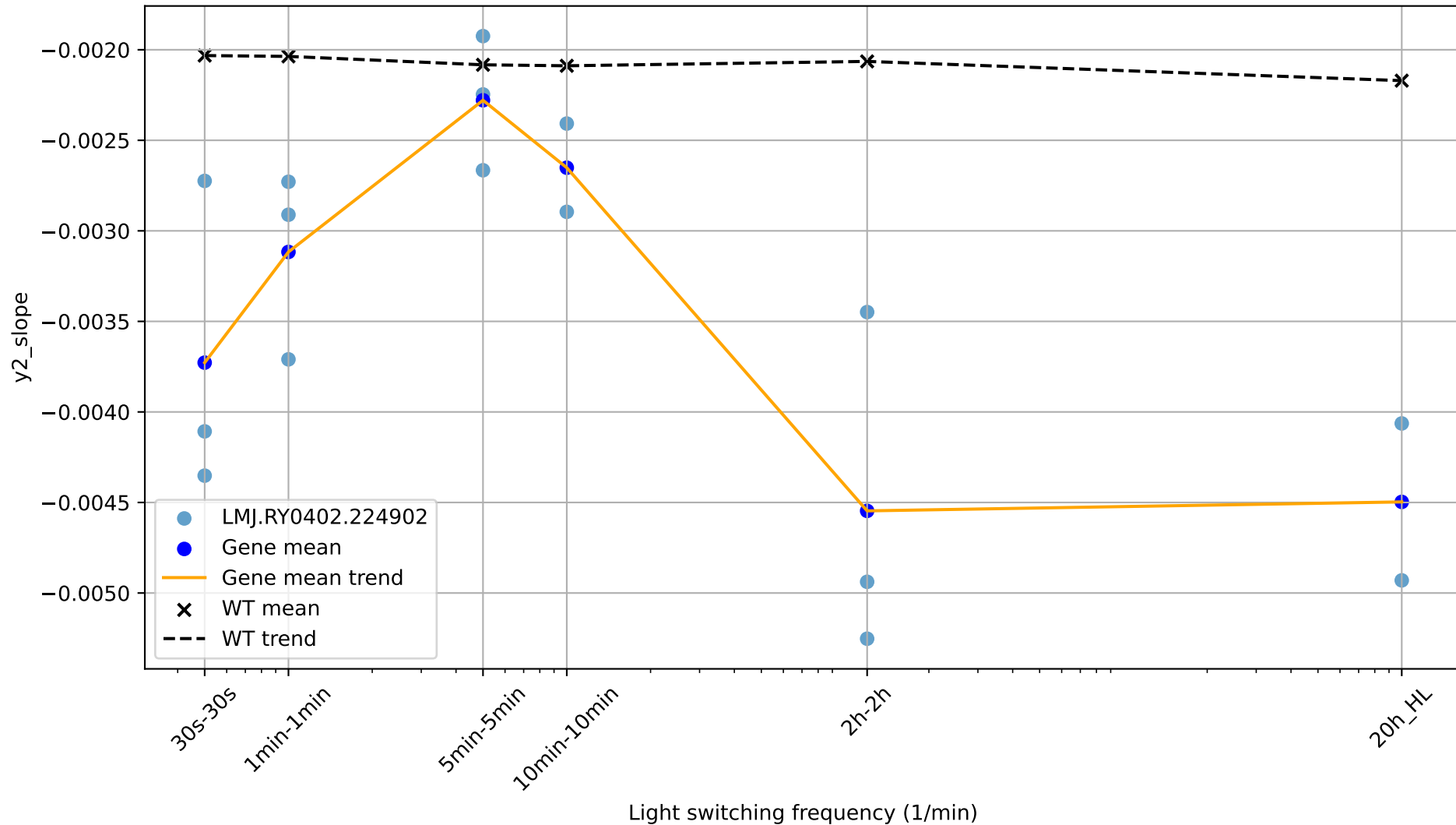
Mutant vs WT - Gene: Cre17.g722000



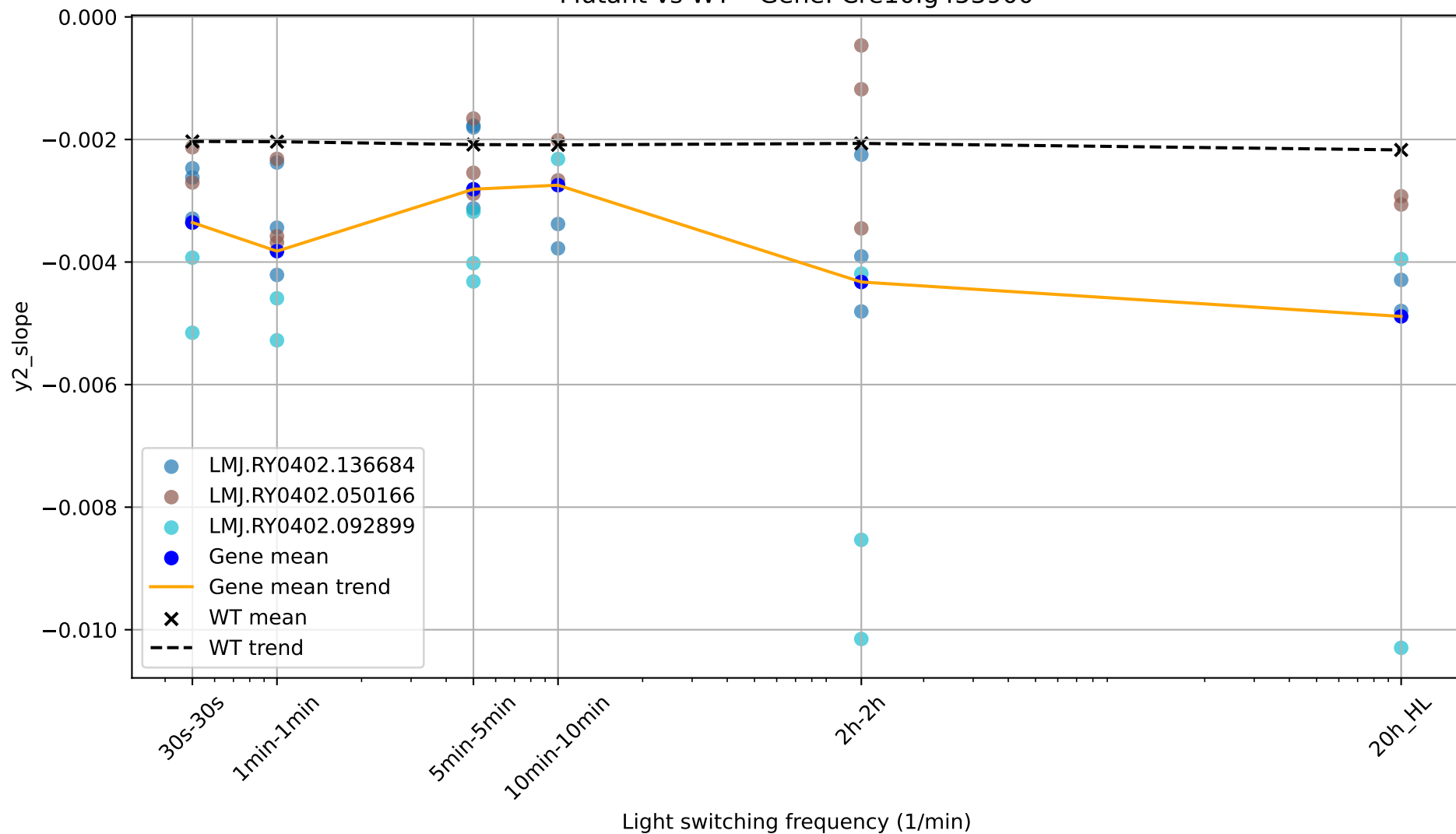
Mutant vs WT - Gene: Cre16.g675350



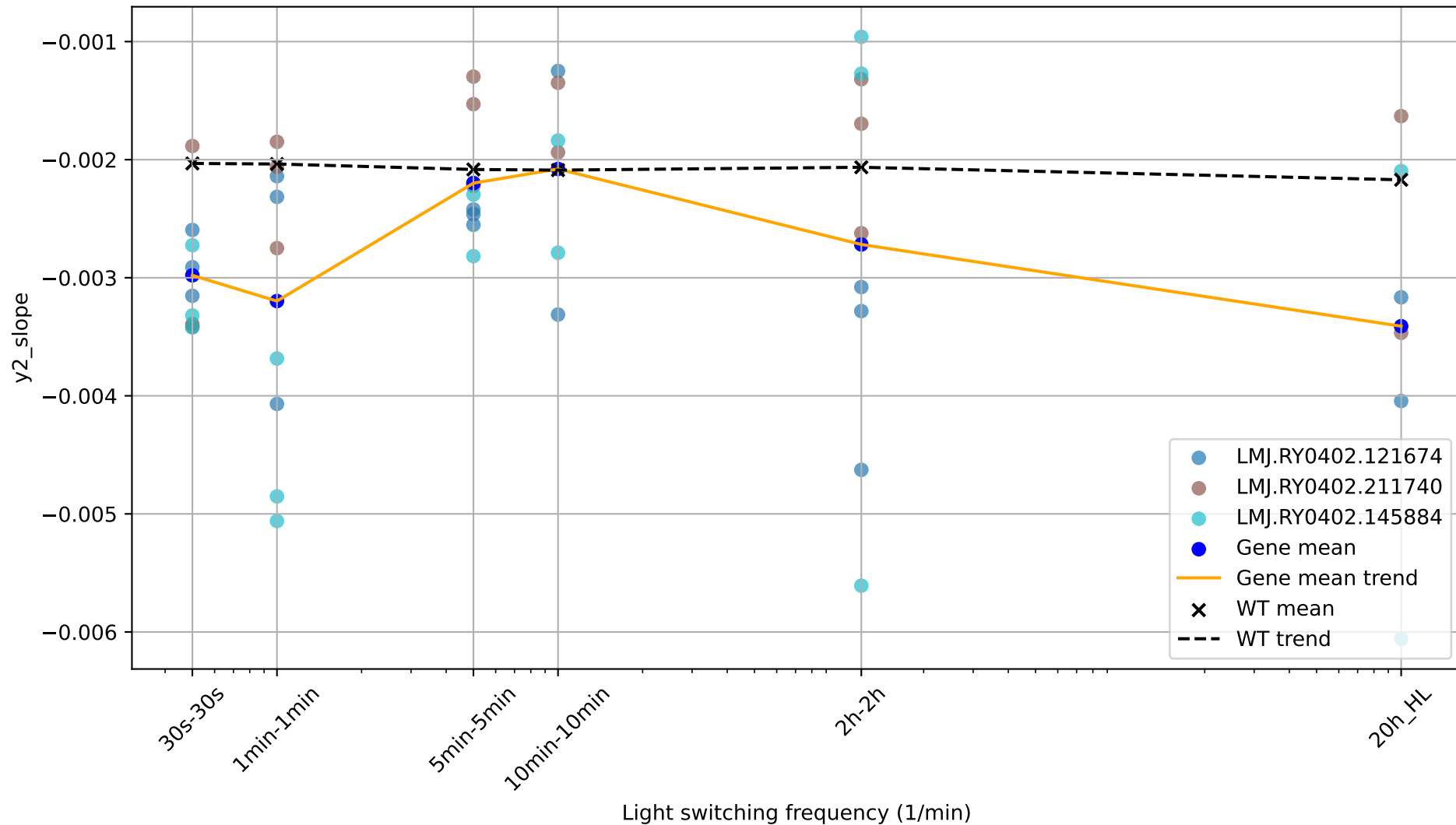
Mutant vs WT - Gene: Cre07.g350250,Cre17.g732533



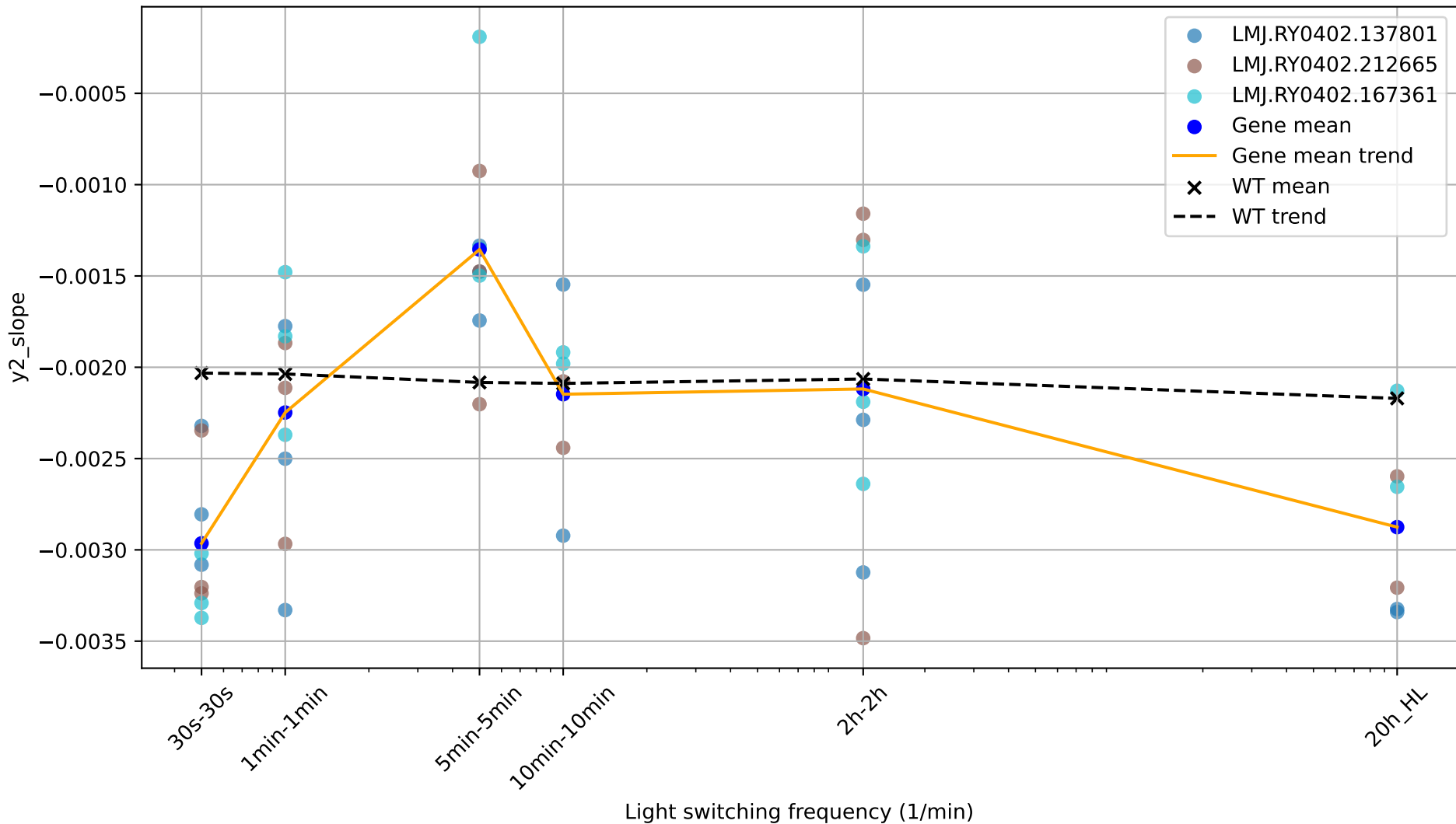
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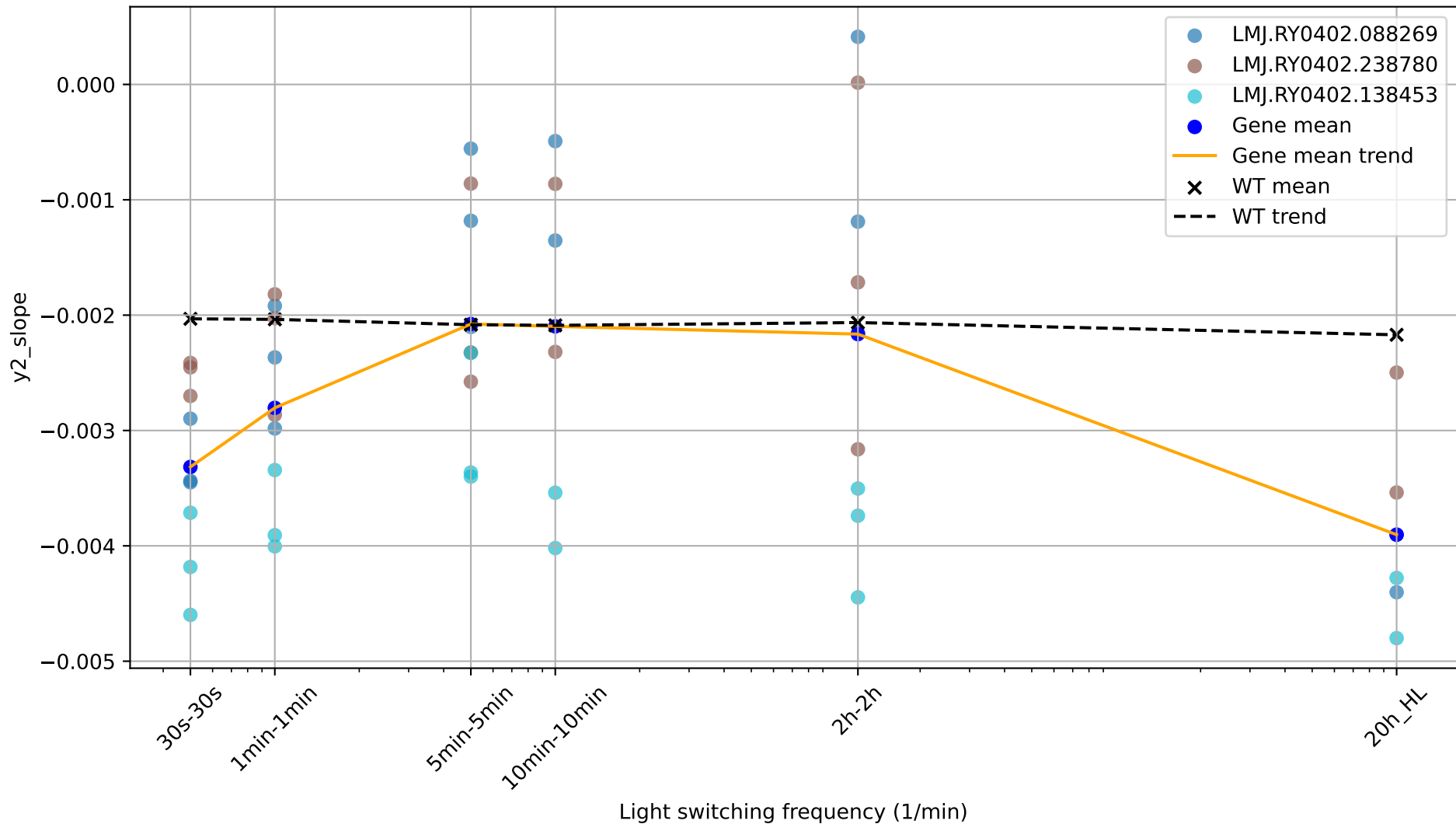
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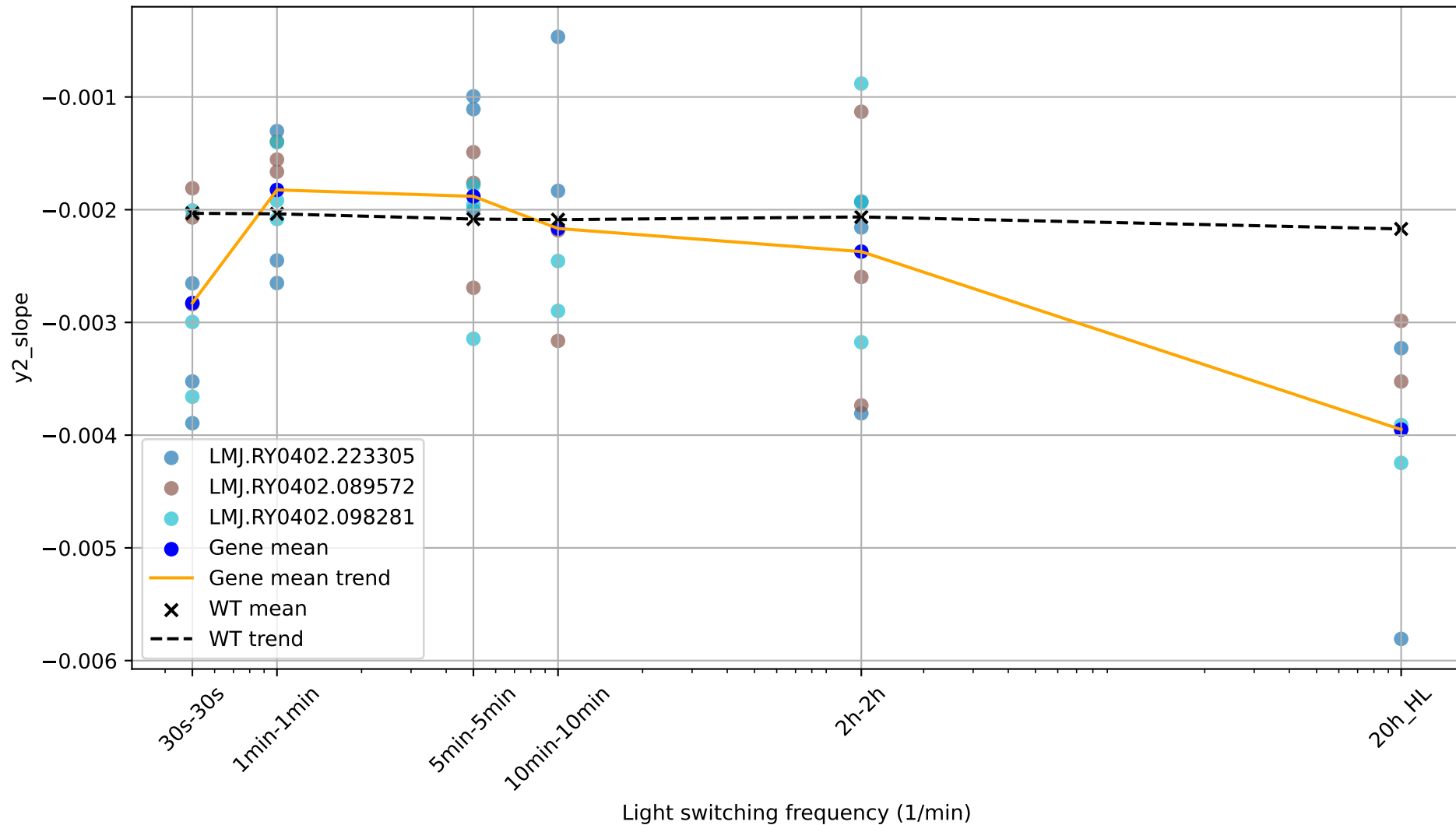
Mutant vs WT - Gene: Cre07.g332901



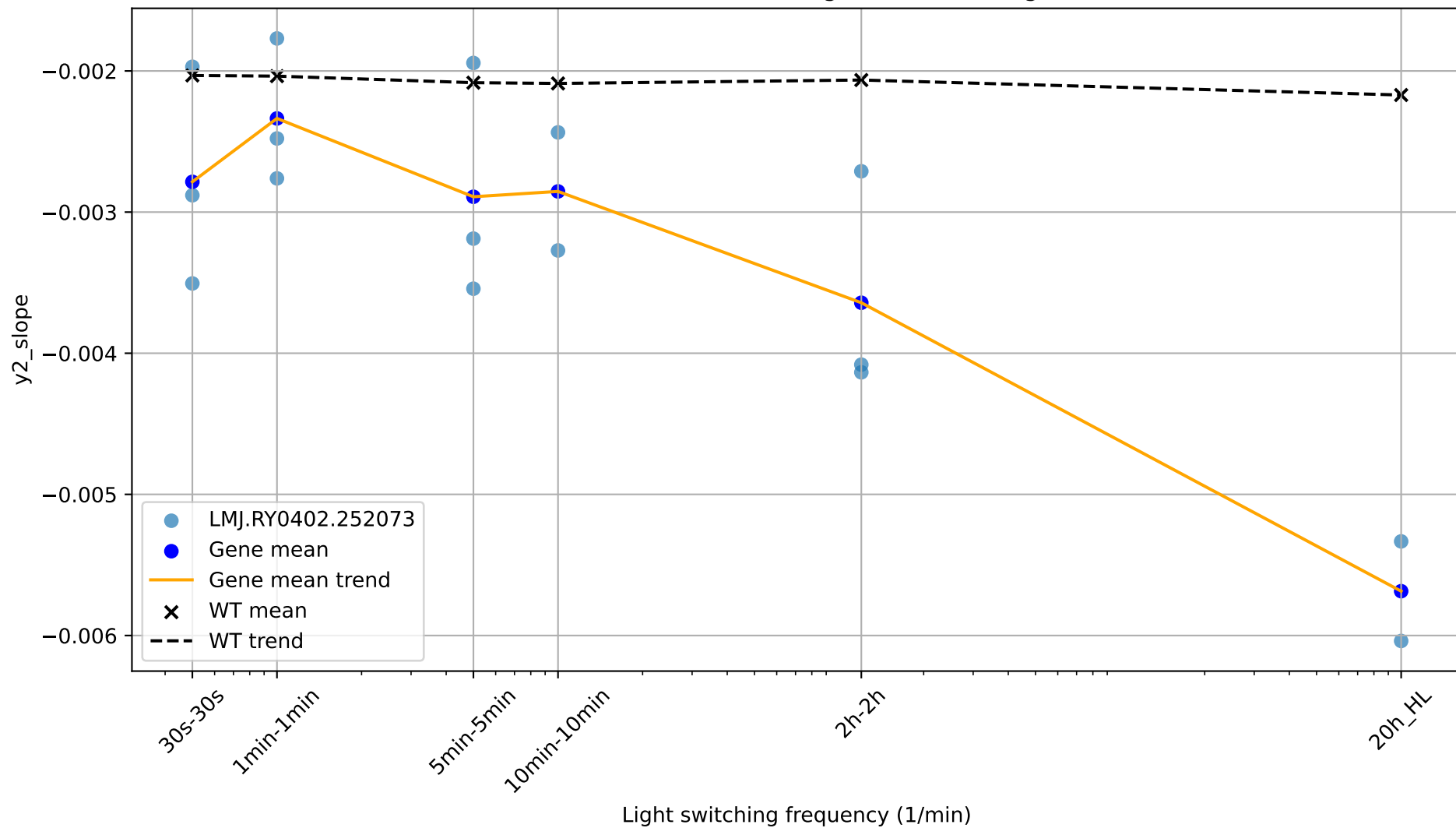
Mutant vs WT - Gene: Cre10.g443150



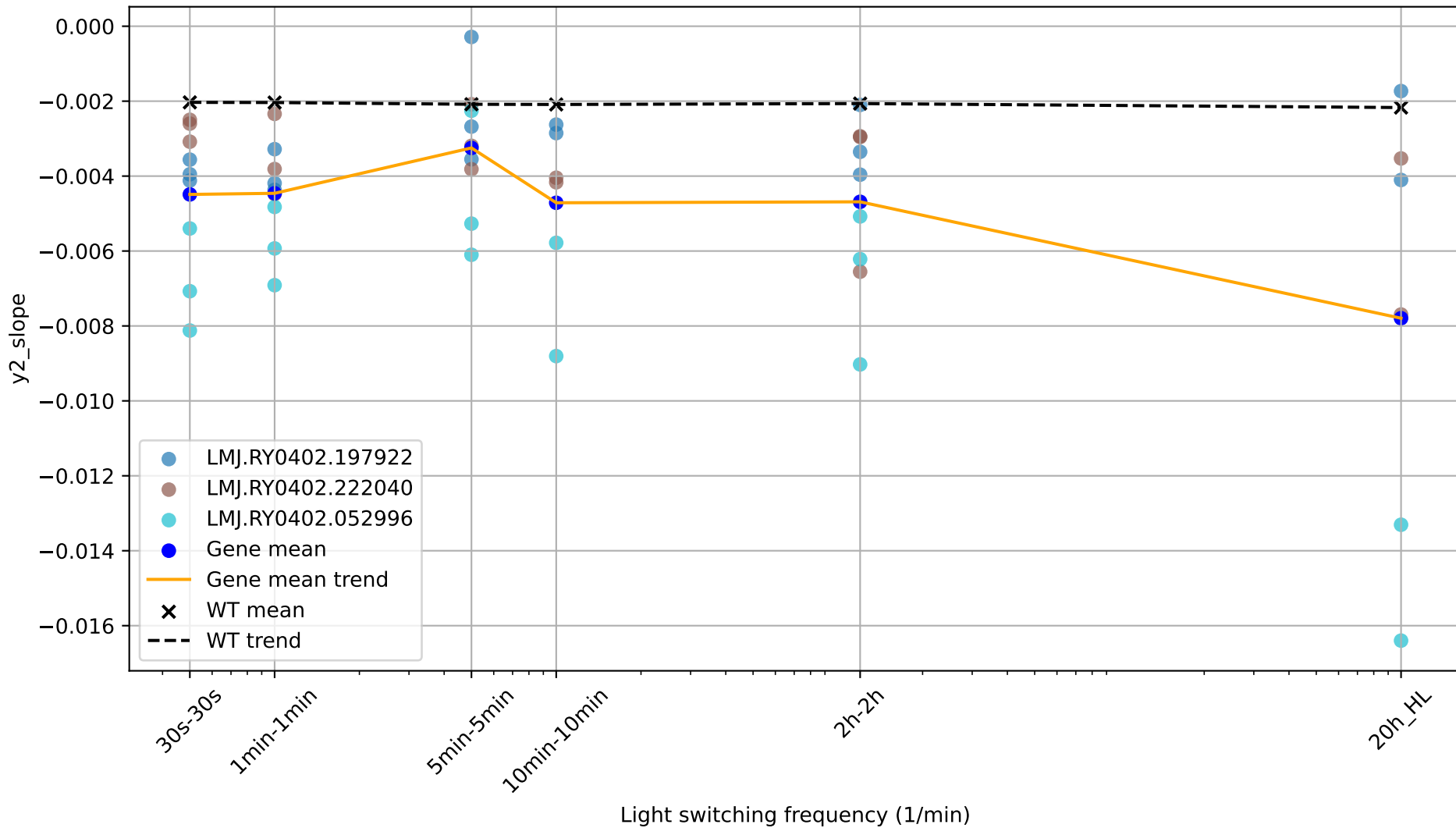
Mutant vs WT - Gene: Cre16.g677200



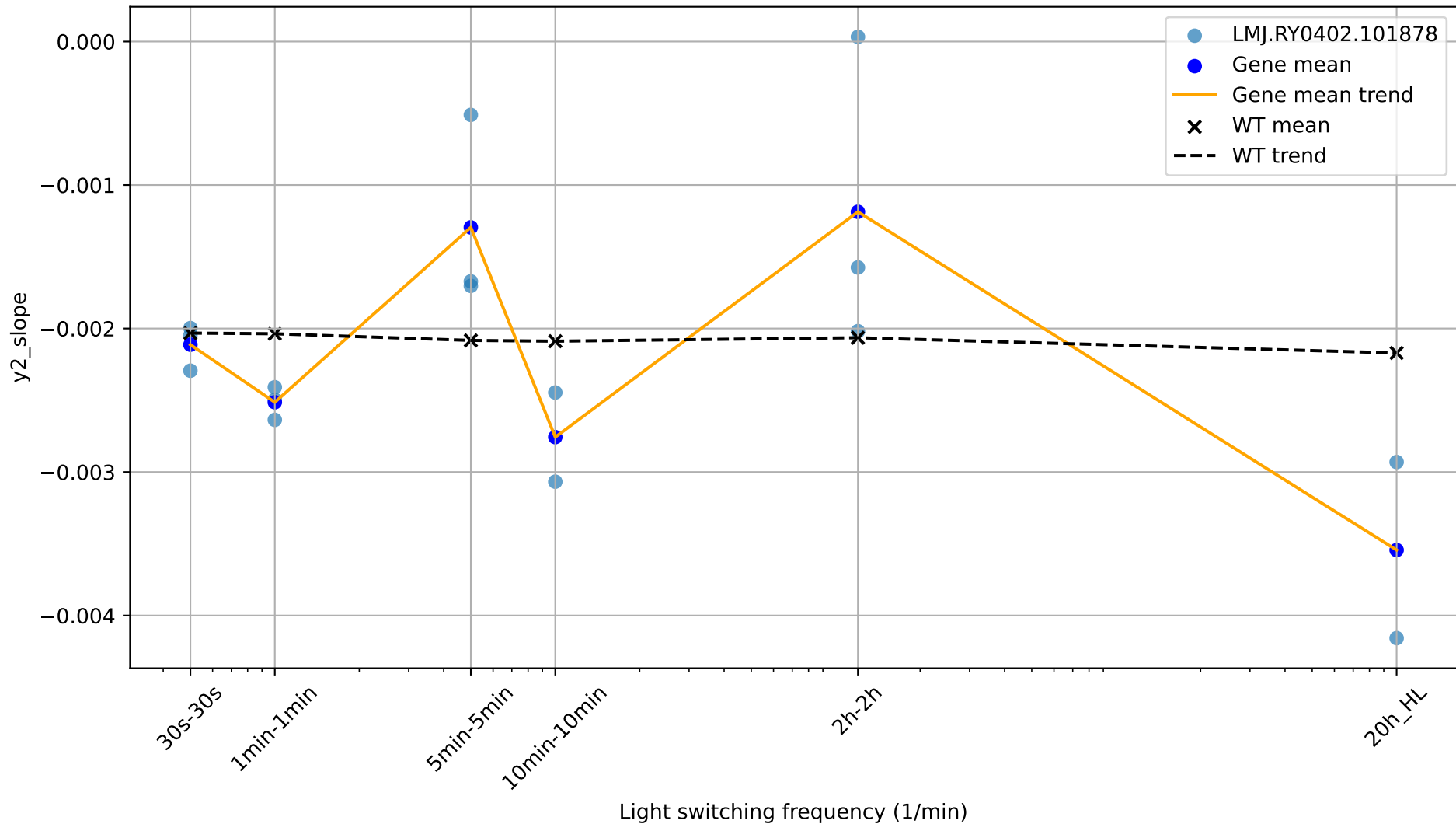
Mutant vs WT - Gene: Cre13.g589450,Cre13.g573500



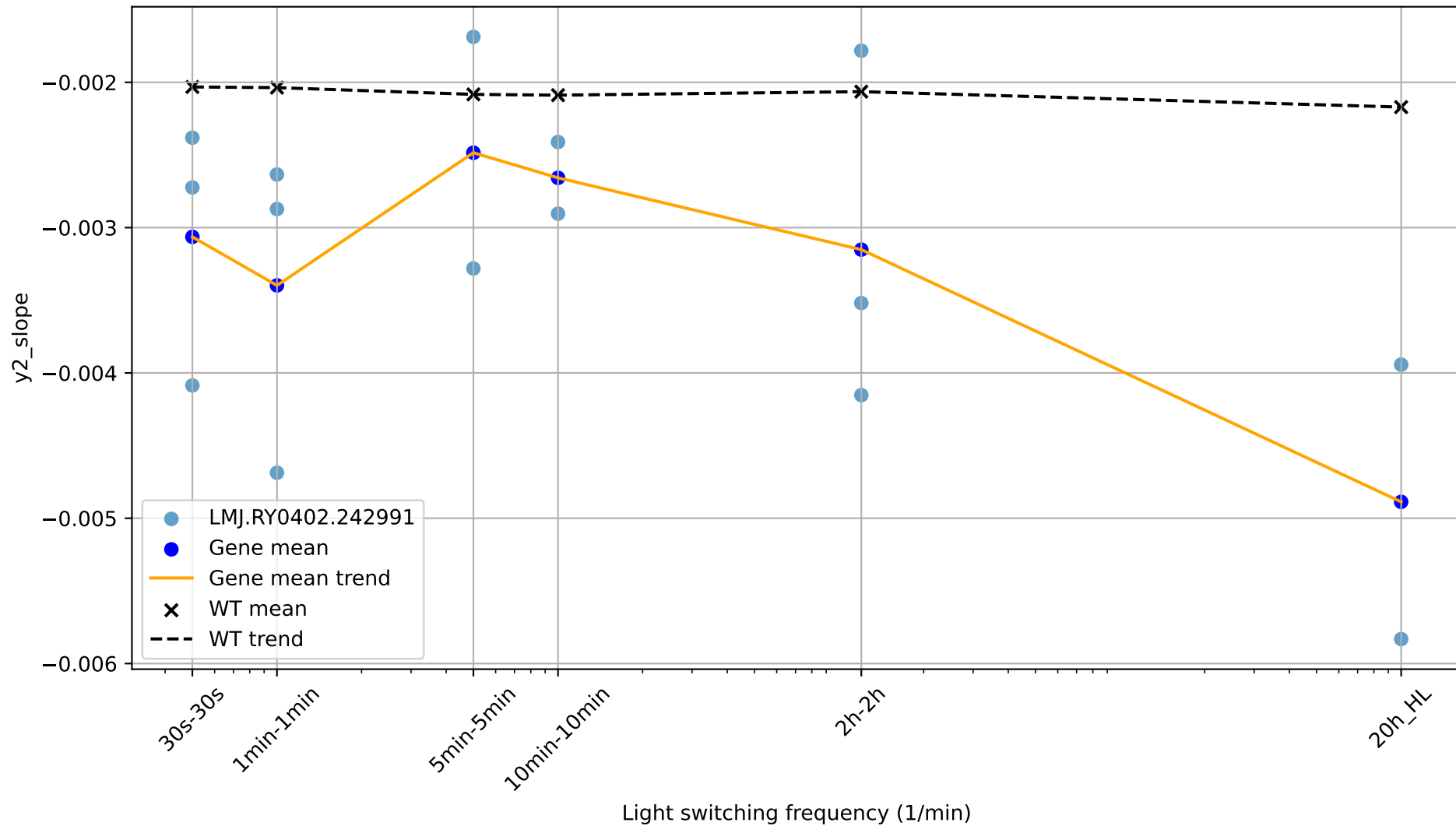
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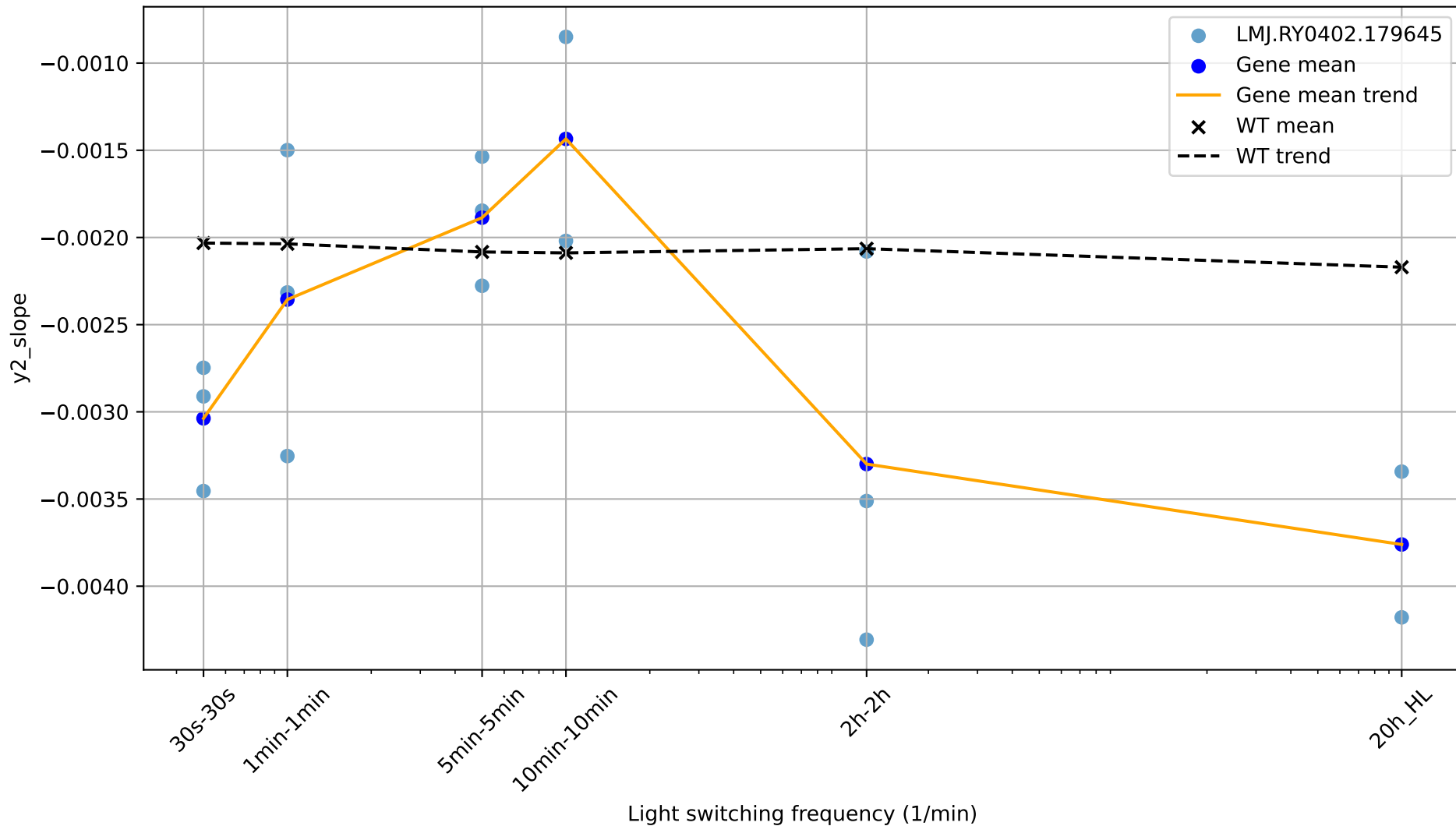
Mutant vs WT - Gene: Cre12.g495175 & Cre12.g495250,Cre12.g495250



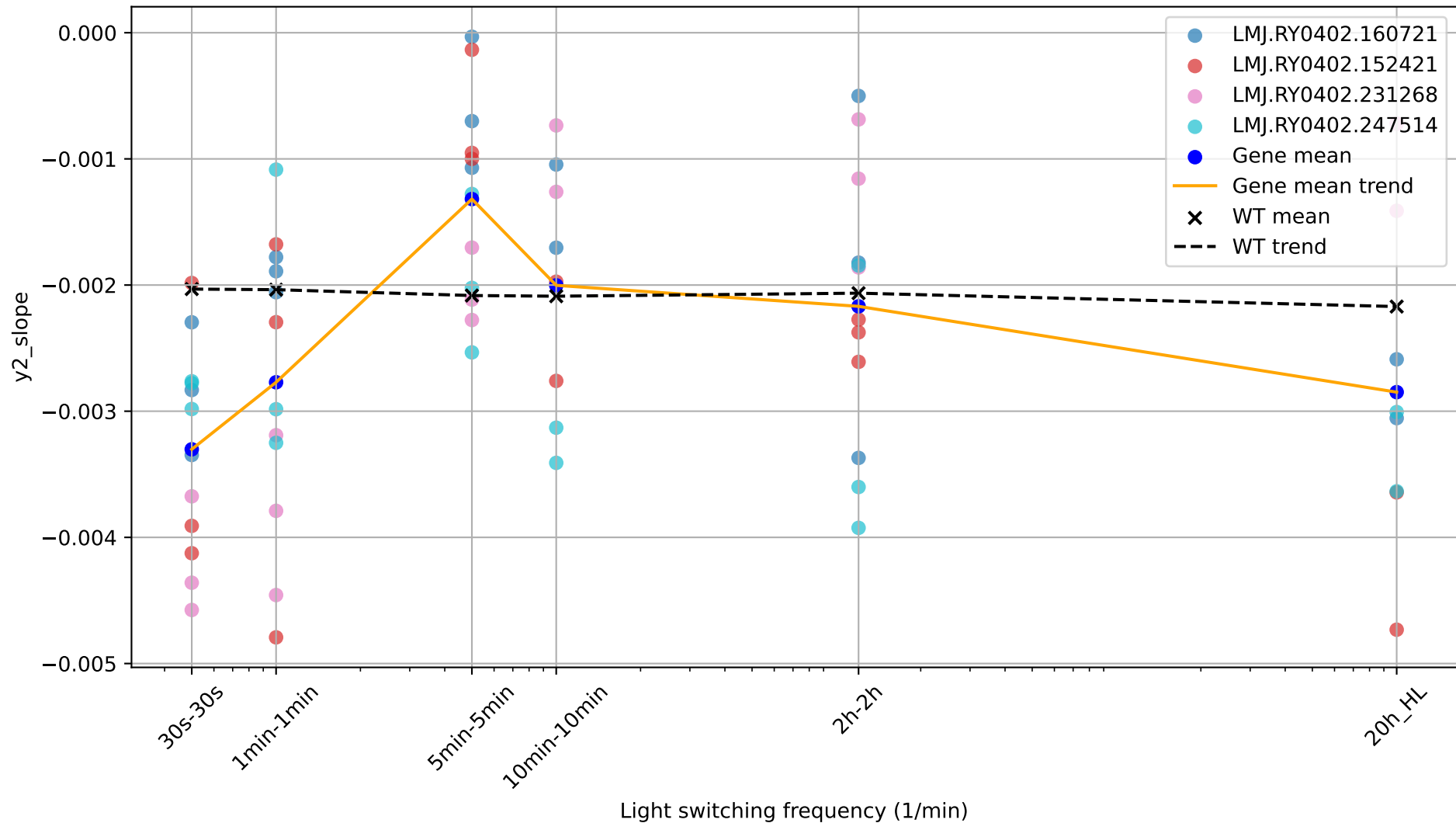
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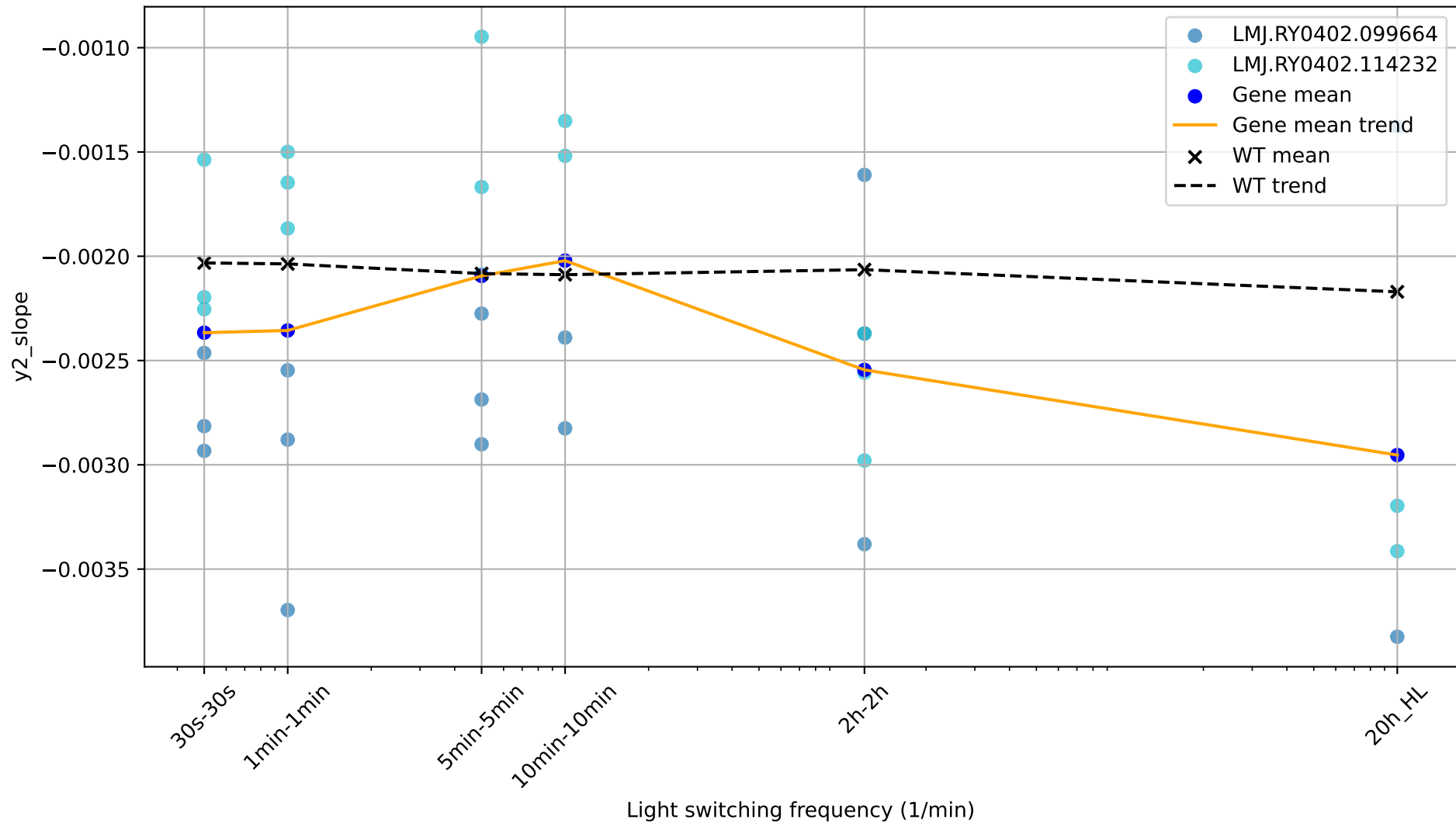
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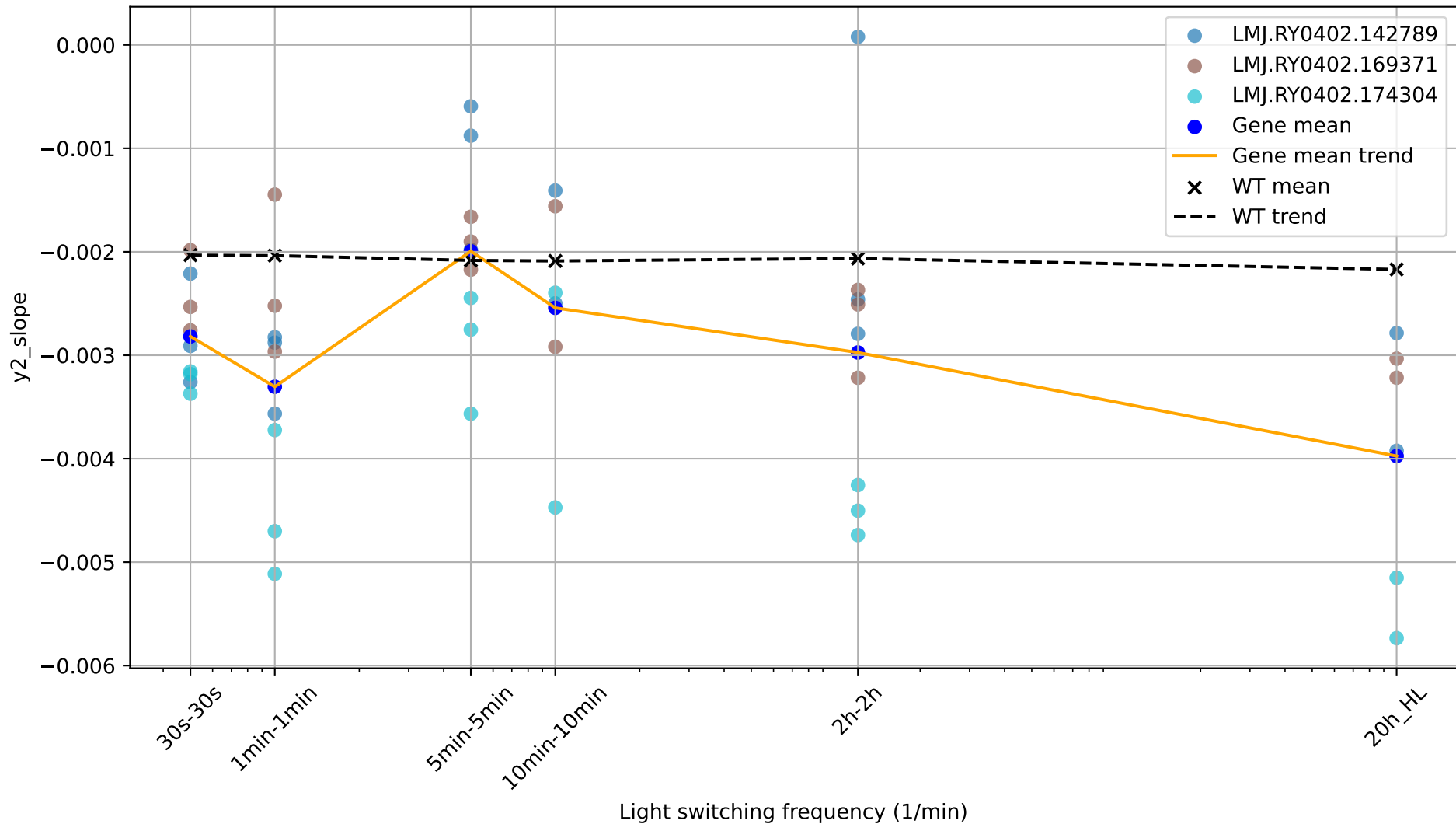
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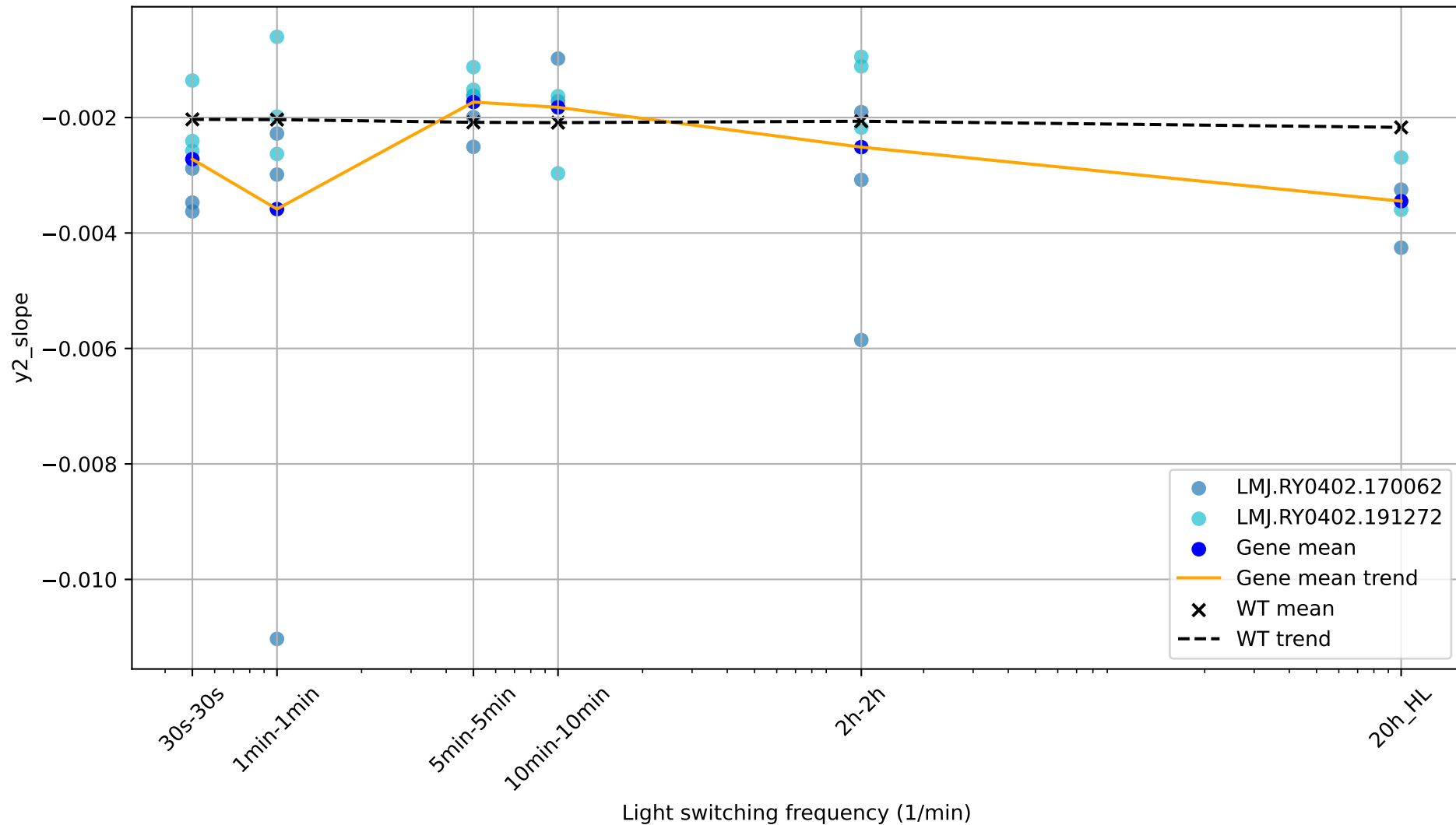
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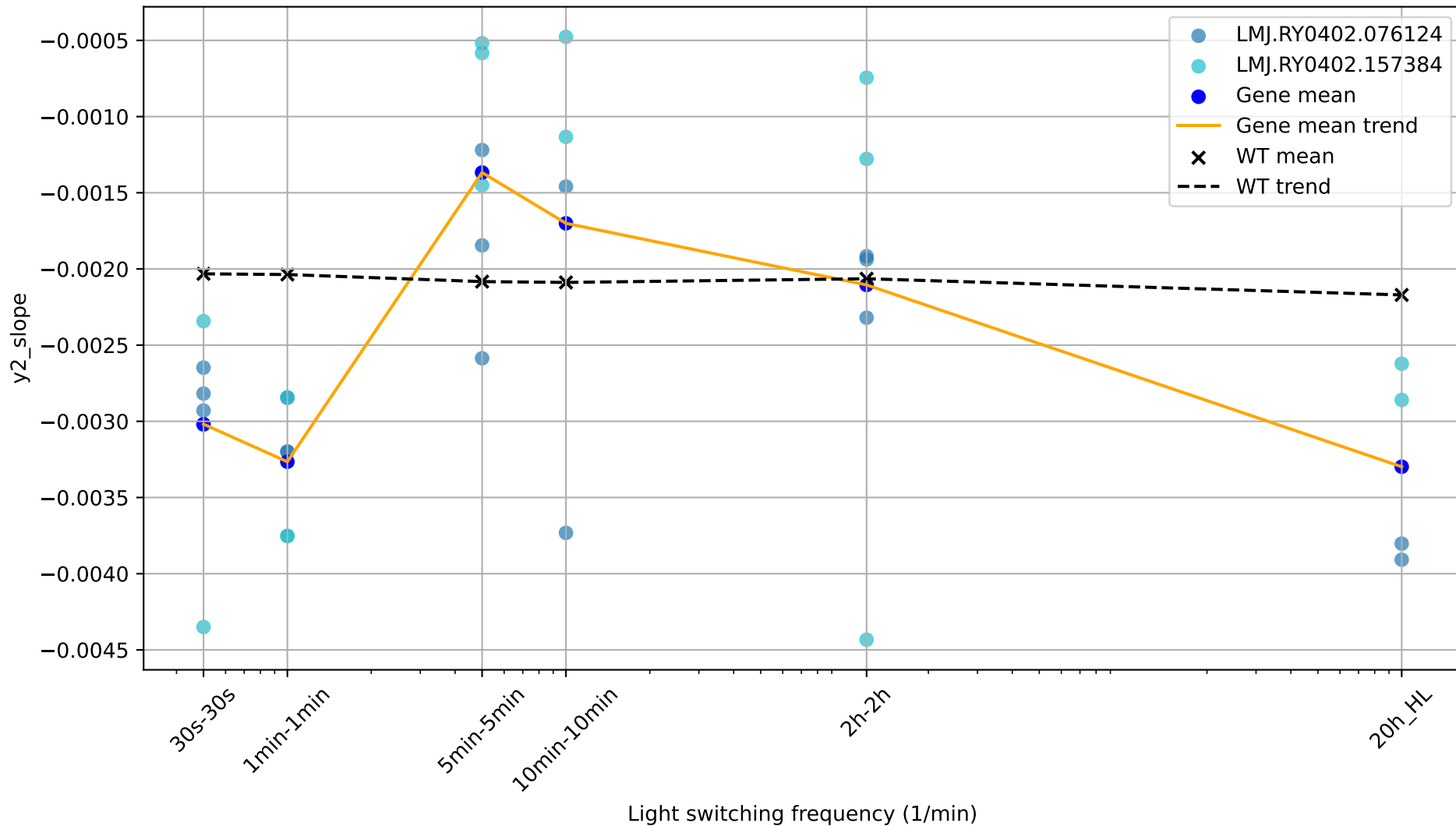
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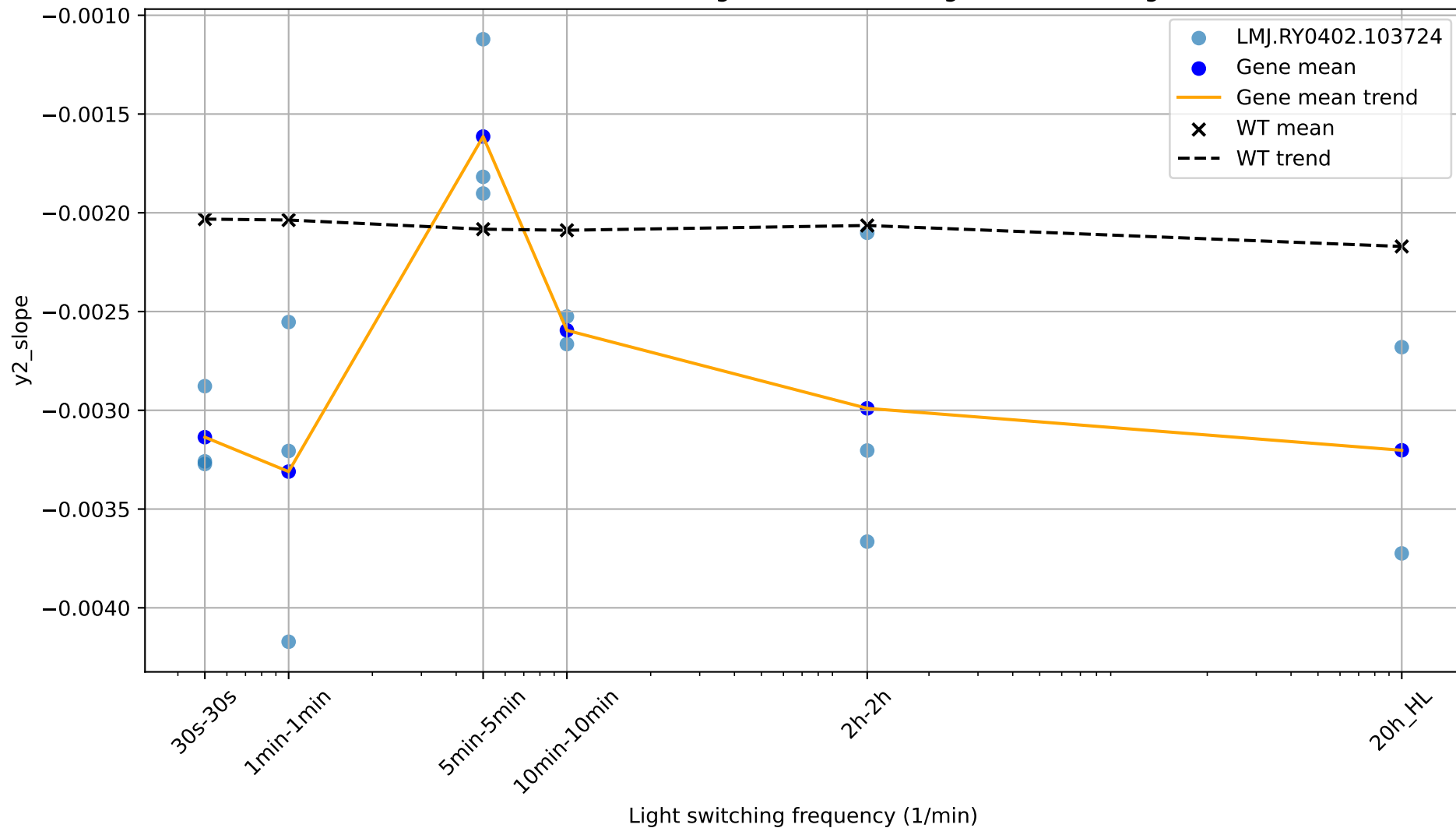
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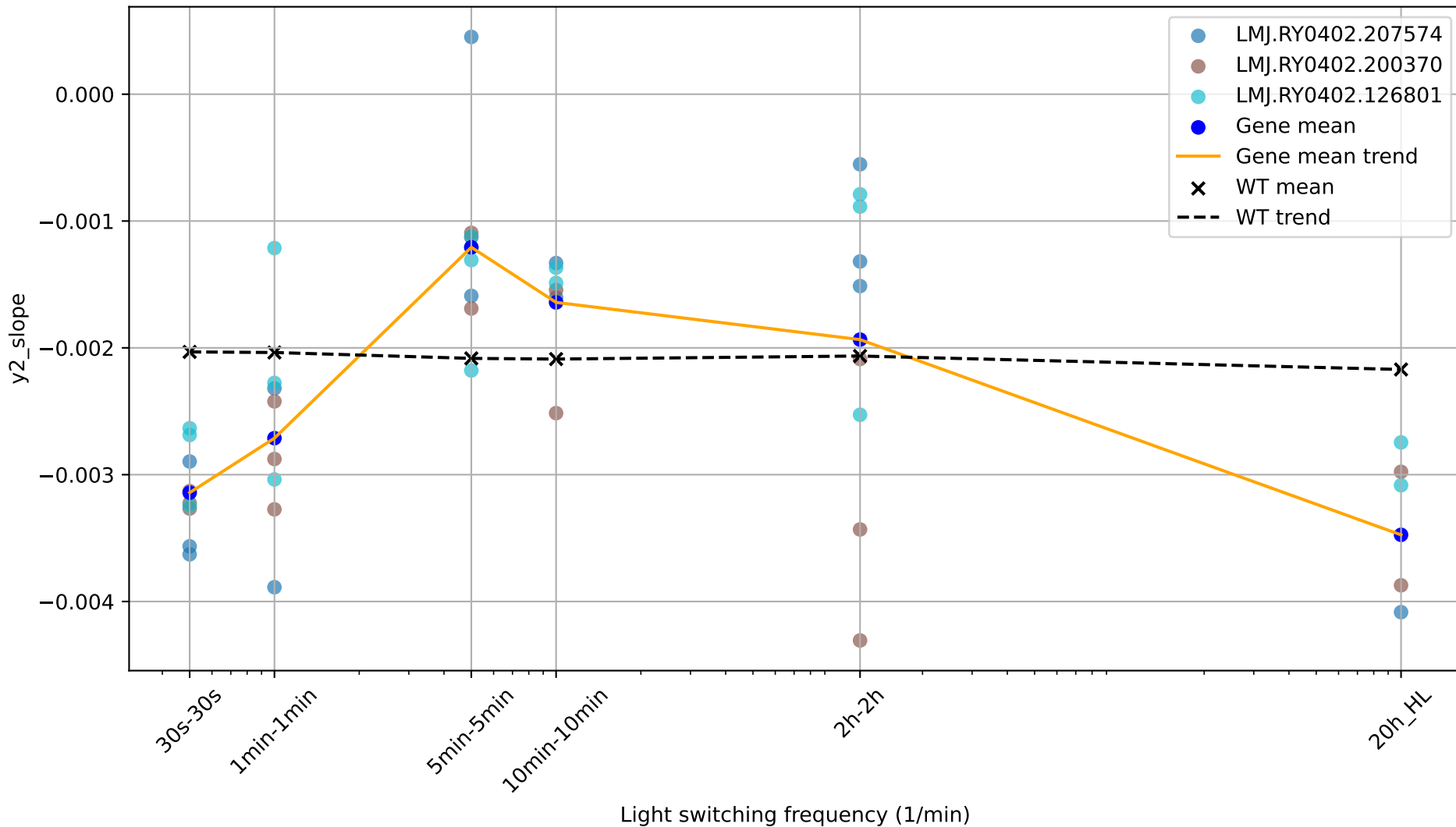
Mutant vs WT - Gene: Cre14.g627576



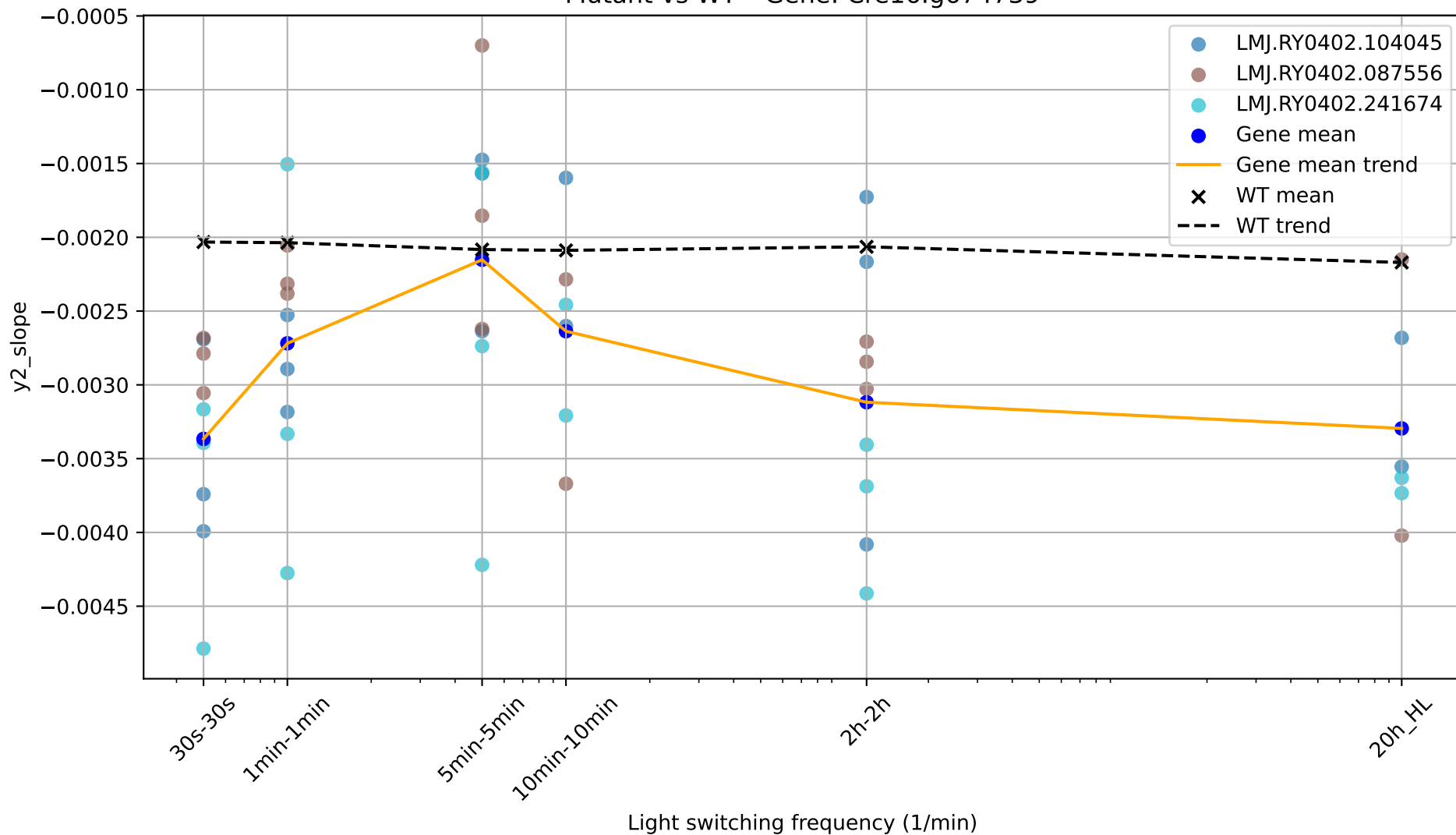
Mutant vs WT - Gene: Cre14.g627251 & Cre14.g627300, Cre14.g627251



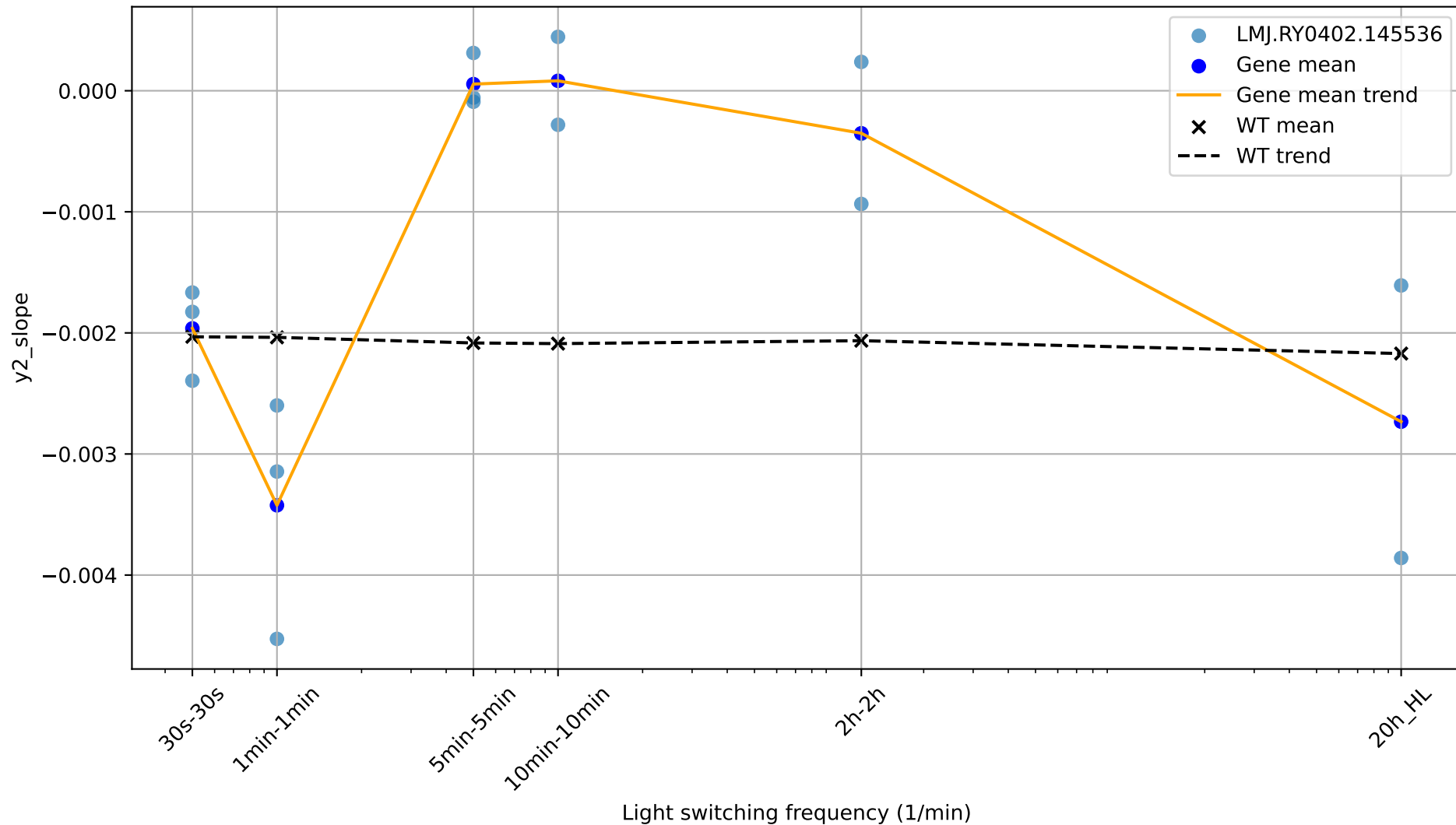
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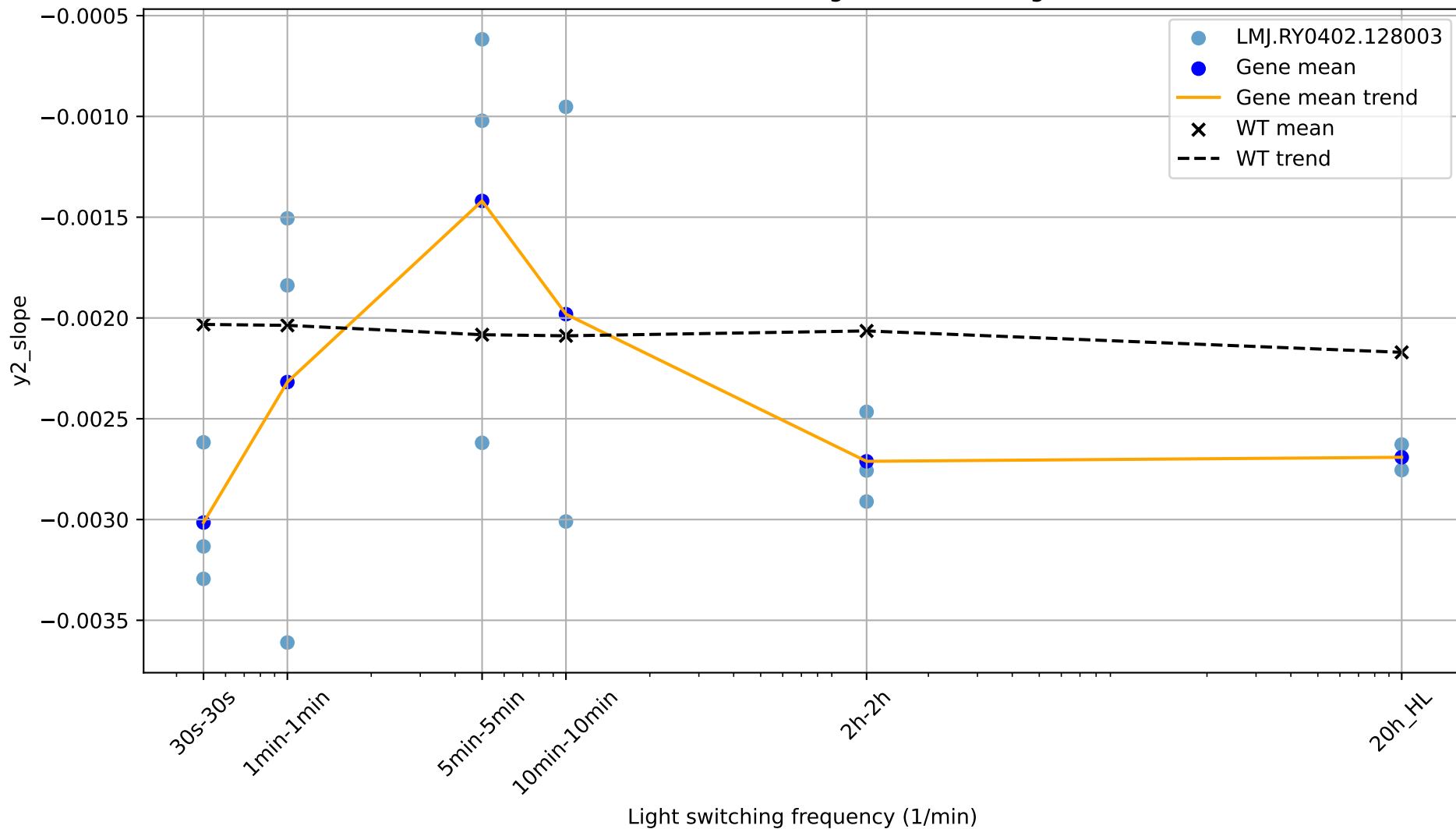
Mutant vs WT - Gene: Cre16.g674739



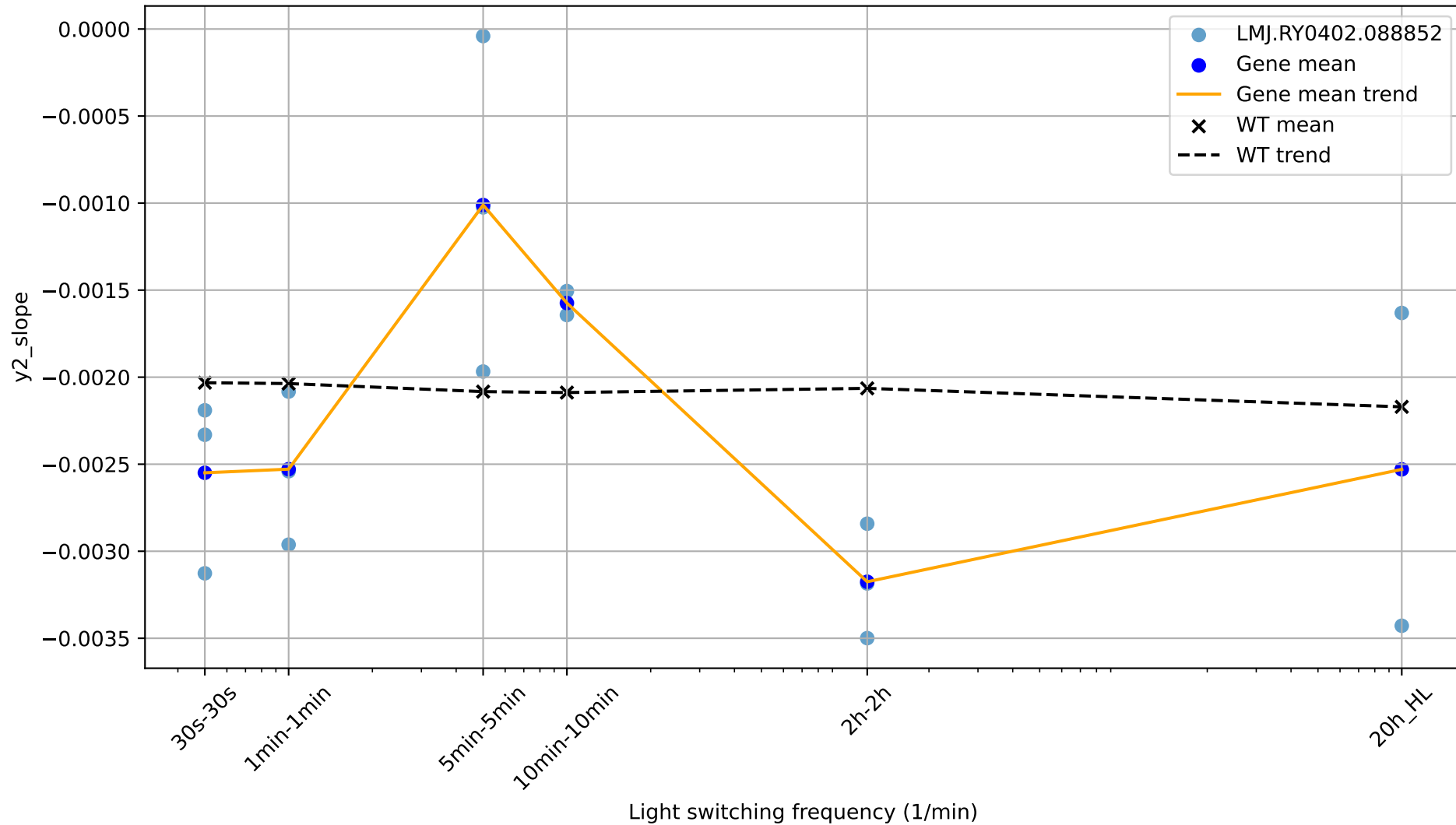
Mutant vs WT - Gene: Cre12.g489300 & Cre12.g489250,Cre05.g236100



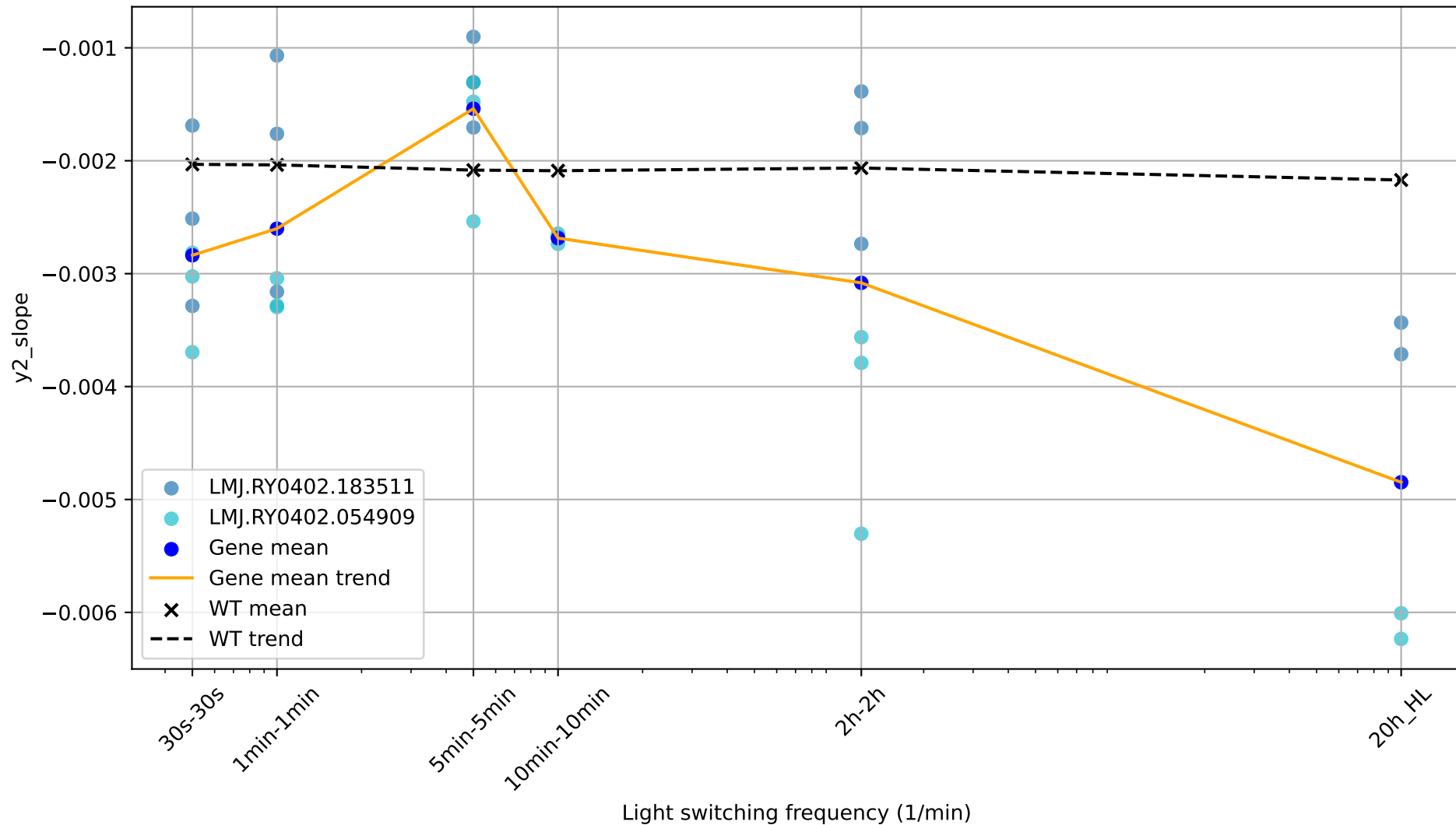
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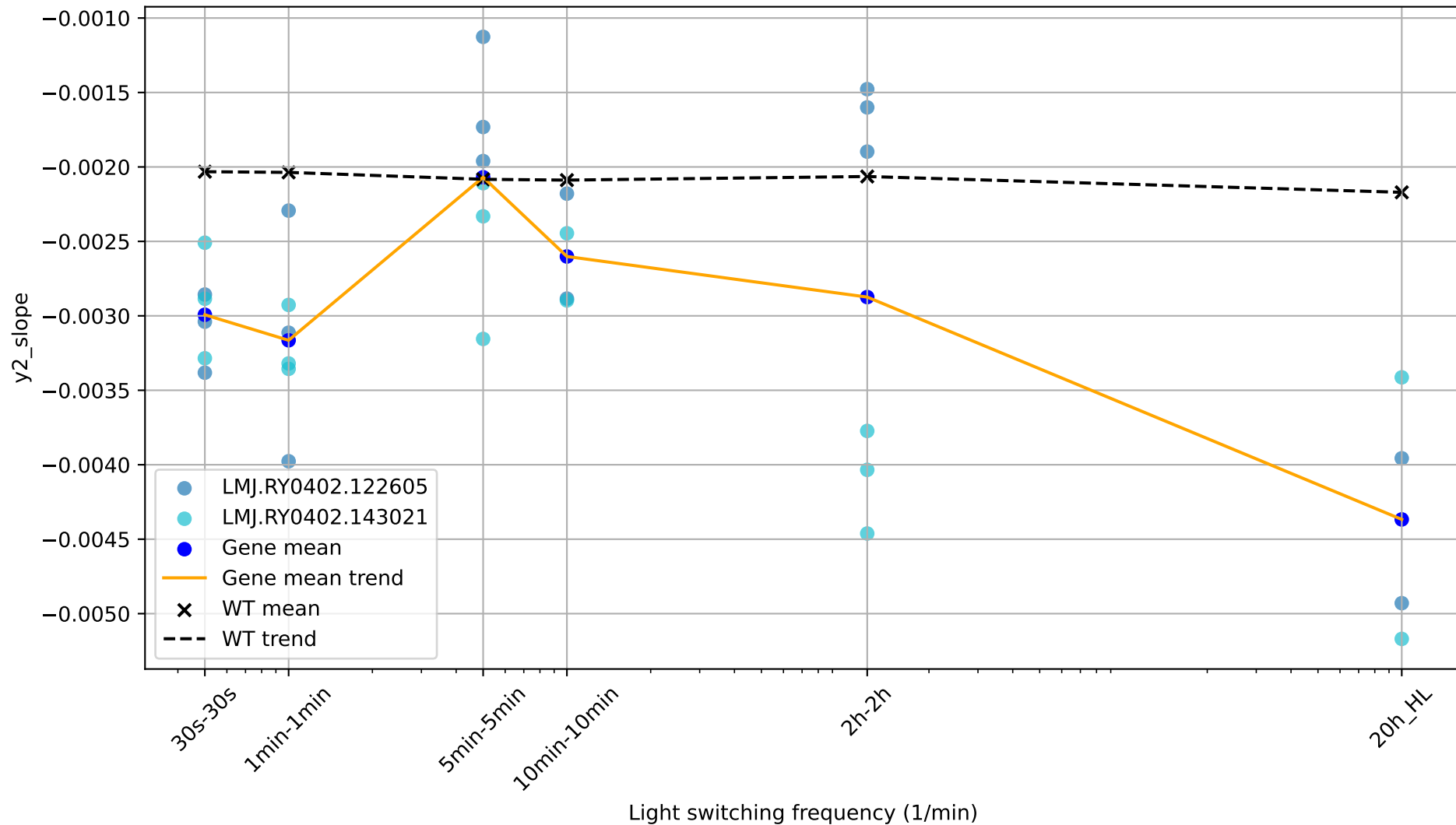
Mutant vs WT - Gene: Cre01.g022650,Cre16.g676869



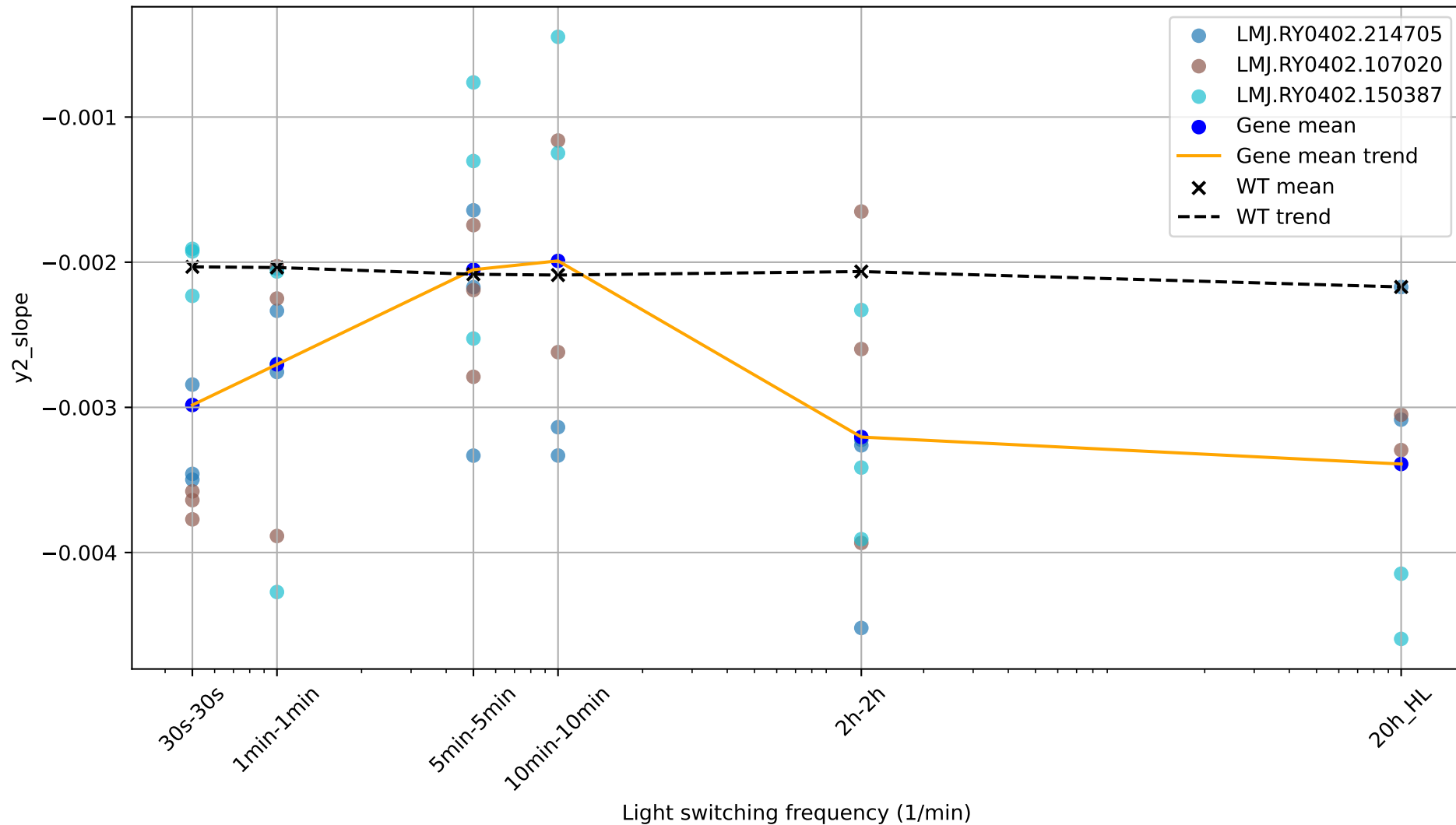
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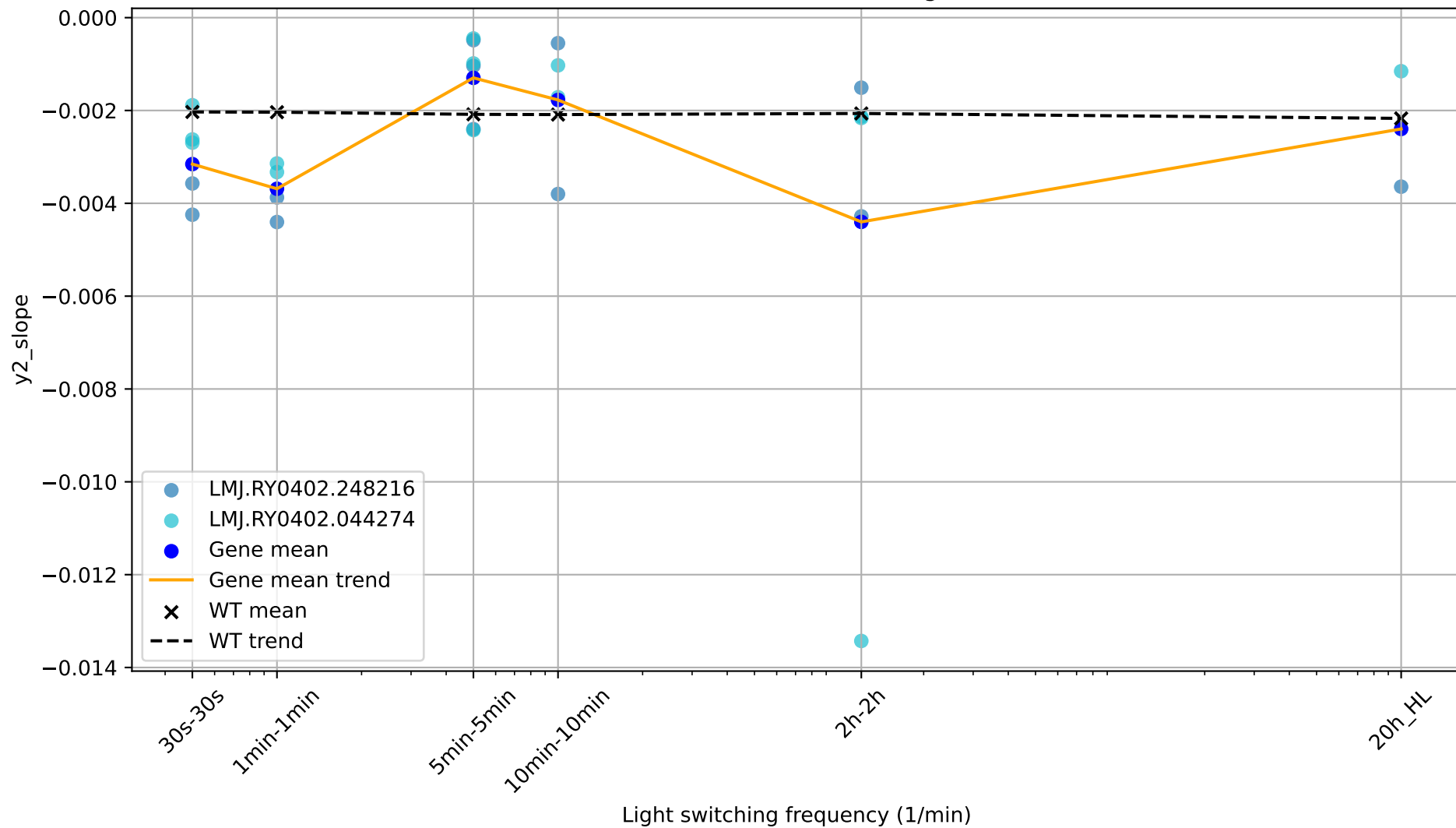
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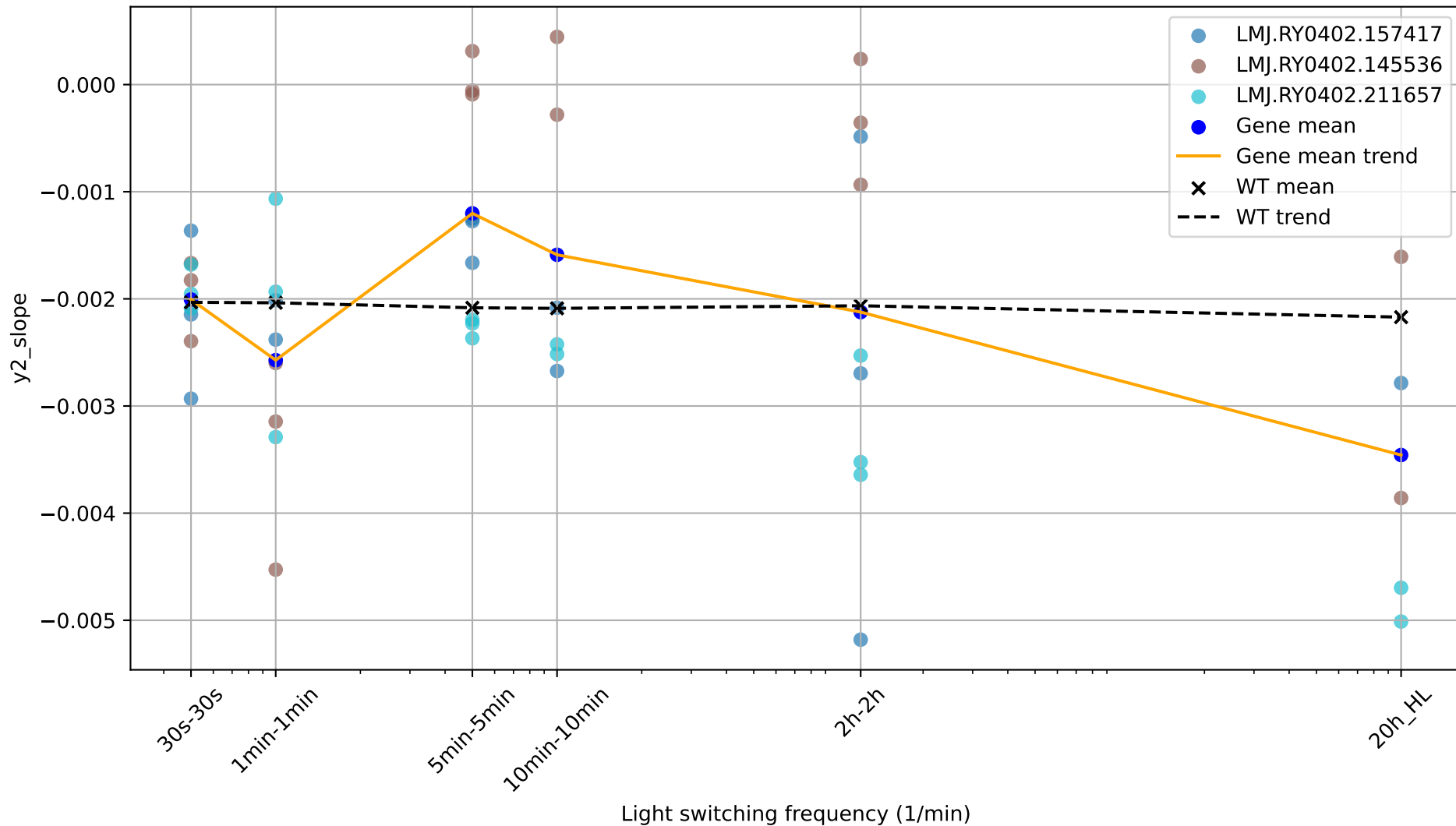
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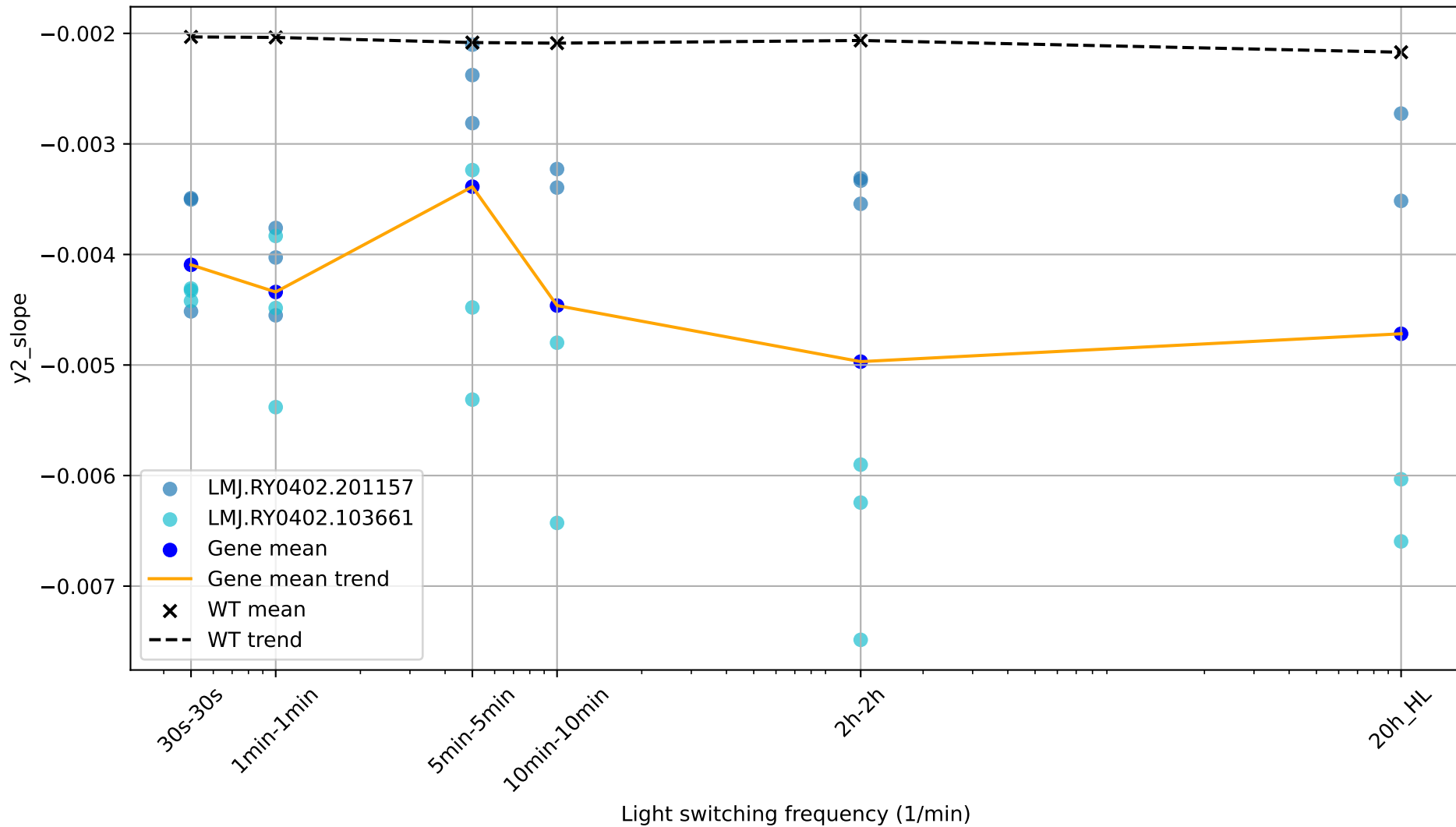
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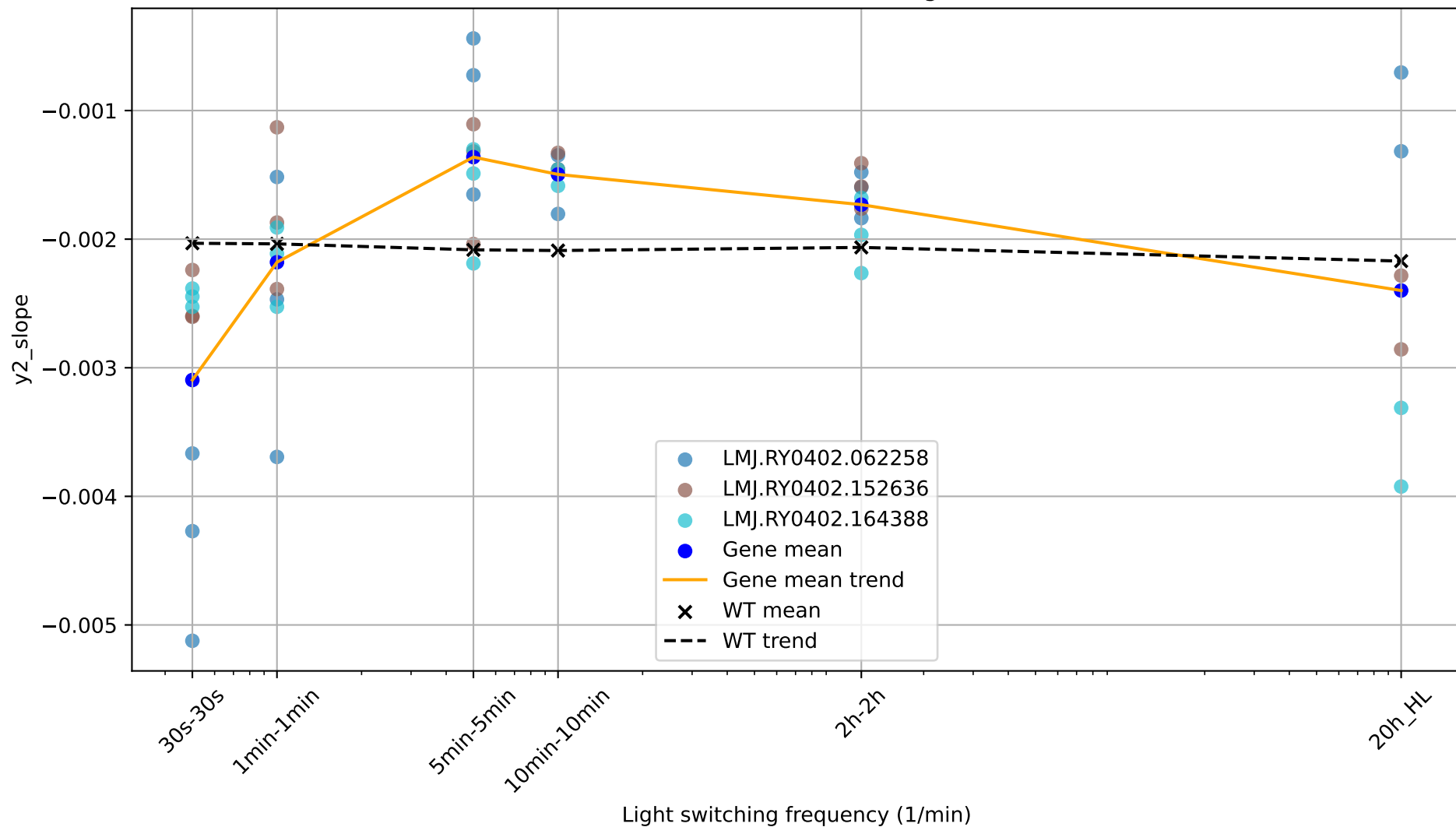
Mutant vs WT - Gene: Cre05.g236100



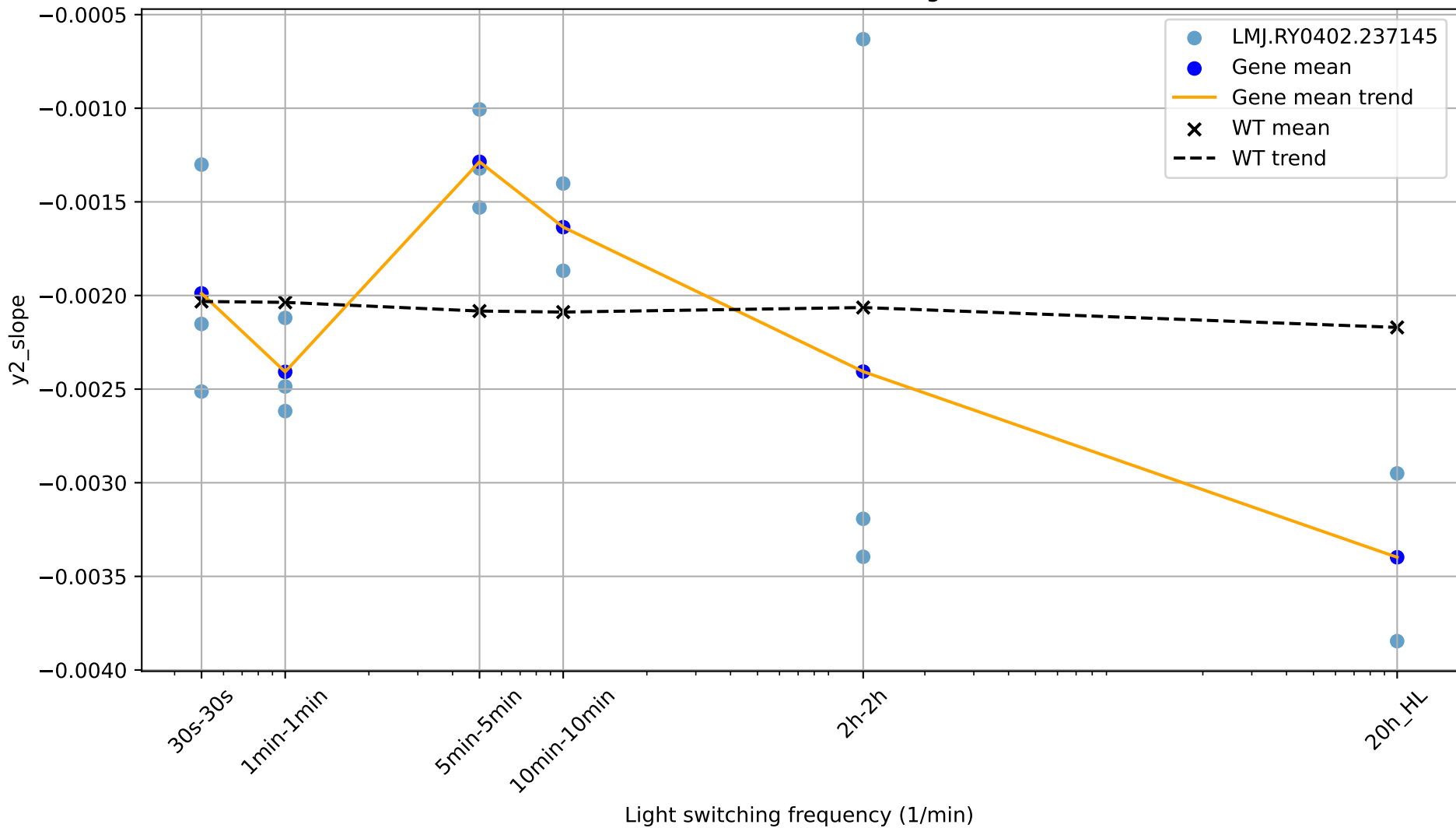
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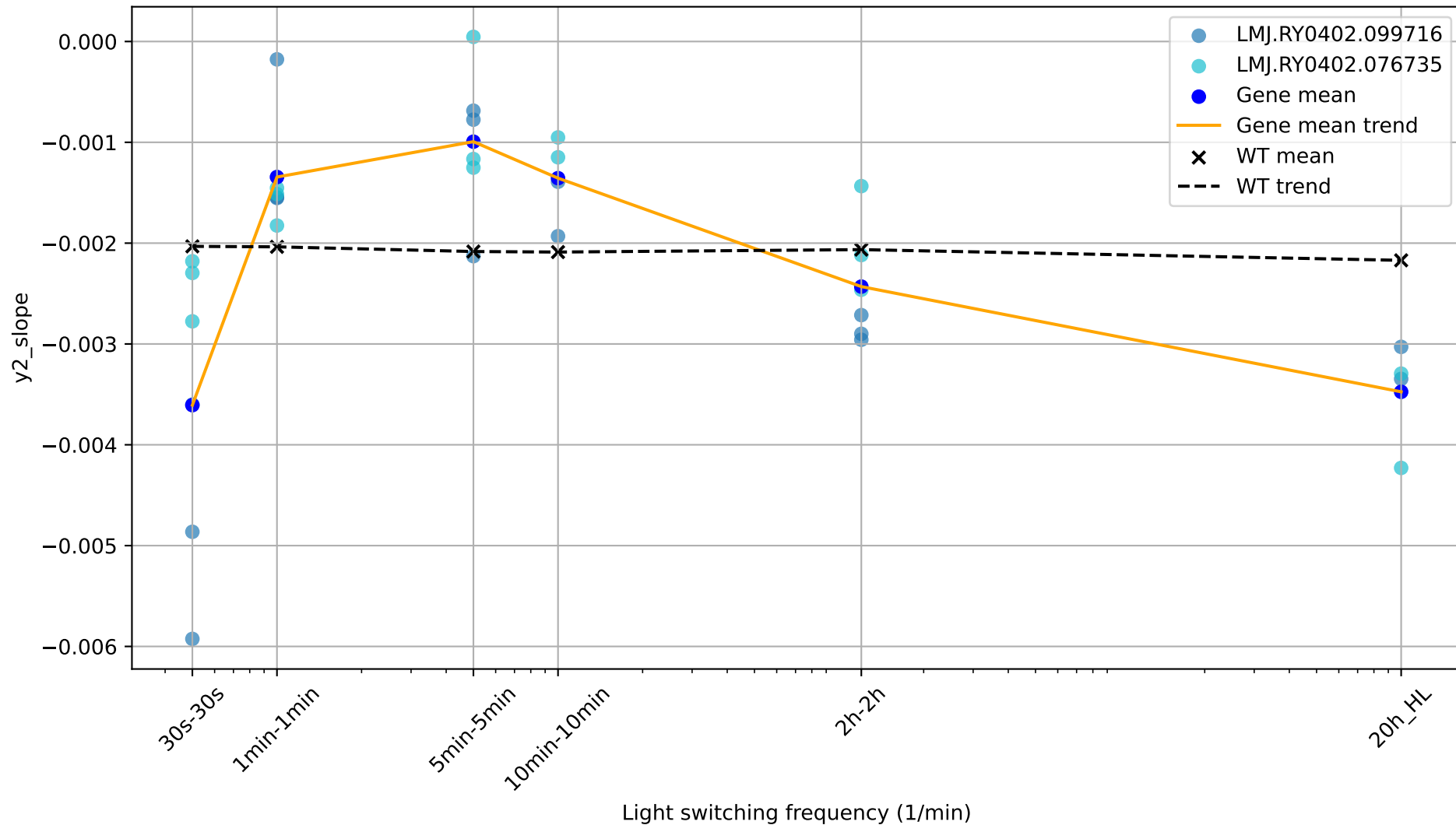
Mutant vs WT - Gene: Cre12.g550800



Mutant vs WT - Gene: Cre17.g710000



Mutant vs WT - Gene: Cre12.g560700



Mutant vs WT - Gene: Cre16.g677200,Cre16.g677250 & Cre16.g677200

